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Historical Data Prediction for Geothermal Systems using Data-Driven Modelling

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Abstract

Geothermal reservoir modelling requires detailed historical extraction data for a given well in order to accurately predict the future health of that well, however this data is rarely available to modellers. Bulk data describing the total extraction rate between multiple wells is available each day, but well-specific measurements are only sparsely recorded. Previously, this data history has been estimated manually, leading to an inaccurate and subjective dataset with unquantifiable uncertainty. This is a serious problem for geothermal reservoir modellers who rely on having accurate data to generate reliable predictions. This report outlines the process undertaken to develop a method of predicting this unknown data history, by inferring the detailed history based on sparse well-specific measurements and dense bulk extraction measurements.

This report introduces two data-driven models to address this issue. Firstly, Loss Function Optimisation (LFO) is developed by designing a loss function to describe the error of a given prediction. Secondly, Linear Least Squares Optimisation (LLSO) builds on this idea by reformulating the system into an inverse problem, and is a much more efficient method of achieving similar results. The final model presented is an optimal Tikhonov-regularised LLSO model which is a computationally efficient, accurate and objective way to predict the historical mass extraction data of a geothermal system. The uncertainty of model predictions can be quantified through Monte Carlo simulation uncertainty propagation.

This model is compared to a Gaussian Process Regression (GPR) model developed in parallel by my project partner, Michael Terekhin, by applying the models to a real geothermal dataset. The final LLSO model can successfully predict the general behaviour of the real geothermal system as the model makes similar predictions to a geothermal expert's estimate. The LLSO model produces a reasonable and smooth estimate with large uncertainty, while the GPR model predictions are highly fluctuating but predict less uncertainty. The LLSO model is also far more efficient to run on a given dataset, making it a viable prospect for use in industry.

Declaration

I hereby declare that:

1. This report is the result of the final year project work carried out by my project partner (see cover page) and I under the guidance of our supervisor (see cover page) in the 2024 academic year at the Department Engineering Science and Biomedical Engineering, Faculty of Engineering, University of Auckland.
2. This report is not the outcome of work done previously.
3. This report is not the outcome of work done in collaboration, except that with a potential project sponsor (if any) as stated in the text.
4. This report is not the same as any report, thesis, conference article or journal paper, or any other publication or unpublished work in any format.

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Date:

12th October 2024

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Nomenclature

$\hat{\Lambda}$	Vector of optimal regularisation parameters from parameter tuning
$\hat{w}$	Matrix of optimal polynomial coefficients for well predictions
λ_1	Parameter weighting the importance given to a TFT measurement, relative to a bulk measurement
λ_2	Regularisation parameter weighting the importance given to first derivative Tikhonov regularisation
λ_3	Regularisation parameter weighting the importance given to second derivative Tikhonov regularisation
λ_b	Balanced parameter weighting giving equal importance to TFT measurements and bulk measurement
Λ	Vector of regularisation parameters for tuning
$\mathcal{L}$	Loss function for unregularised AHOP LFO model
$\mathcal{L}_1$	Loss associated with bulk measurements
$\mathcal{L}_2$	Loss associated with TFT measurements
$\mathcal{L}_c$	Loss function for constant LFO model
$\mathcal{L}_l$	Loss function for linear LFO model
$\mathcal{L}_{ave}$	Average total loss of LFO model
$\mathcal{L}_{Ridge}$	Loss function for ridge-regularised AHOP LFO model
$\mathcal{L}_{Tik}$	Loss function for first and second derivative Tikhonov-regularised AHOP LFO model
A	Model and data matrix for LLSO model
A_1	Data matrix for LLSO model
A_2	Model matrix for LLSO model
$A_{1,Tik}$	Data Matrix with finite difference stencils for first and second derivative Tikhonov regularisation
$A_{Tik,1}$	Matrix of finite difference stencils for first derivative Tikhonov regularisation
$A_{Tik,2}$	Matrix of finite difference stencils for second derivative Tikhonov regularisation
b	Vector of constants from differentiating the loss function
D	Total number of TFT points provided to the model
d	Matrix of whether a TFT point is provided for each well across time
e	Vector of unknown noise for LLSO model
E_{ave}	Average error of prediction at a single well and timepoint
i	Subscript identifying the well number

k	Highest degree of arbitrarily high degree polynomial
L	Matrix of coefficients from differentiating the loss function
M	Matrix of bulk mass extraction data across time
m	Matrix of true mass extraction data from each well across time
n	Number of wells in the geothermal system
q	Matrix of well predictions for all wells at all timepoints for LLSO model
T	Number of discretized timepoints in the geothermal system
t	Subscript identifying the timepoint
w	Vector / matrix of polynomial coefficients for LFO model
w_c	Vector of y-intercepts of well prediction equations for linear LFO model
w_g	Vector of gradients of well prediction equations for linear LFO model
$W_i(t)$	Polynomial equation for well i as a function of time t
x	Vector of polynomial coefficients for LLSO model
y	Vector of observed data for LLSO model
z	Matrix of whether each well was on or off across time

Chapter 1

Introduction

1.1 Problem Background

Geothermal energy is a renewable and environmentally-friendly energy source most commonly found on boundaries between different tectonic plates[1], such as New Zealand. Geothermal reservoirs hold water (in liquid and gaseous forms) at high temperatures and pressures in underground volumes of permeable rock. This hot steam can be extracted to provide physical heat to a town or broader area, or it can be used to generate electrical energy by driving large steam turbines [2]. The electricity generated by this can then be fed into the electrical grid, which provides the country with a clean, renewable electricity.

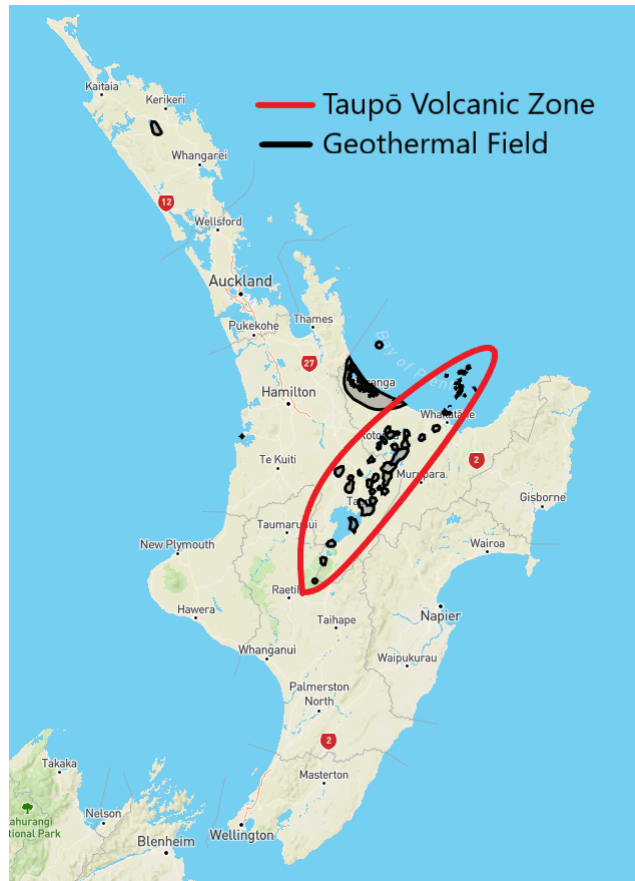


Figure 1.1: Geothermal fields and Taupō Volcanic Zone in New Zealand's North Island. Adapted from [3].

New Zealand's North Island has an abundance of Geothermal reservoirs [3]. The majority of these lie in the Taupō Volcanic Zone, which runs Southwest from Whakaari (White Island) through Lake Taupō, to Mount Ruapehu, as shown in Fig. 1.1. New Zealand is thought to have 1300 to 2500 MWe of available geothermal energy [4]. It is not possible

to artificially create more geothermal reservoirs, so we must be sure that we are using these geothermal resources responsibly. Because of the numerous benefits of geothermal electricity generation, it is important to invest in geothermal energy and ensure that we are properly caring for our geothermal resources. It is important not to extract too much mass from a geothermal reservoir because over-extraction may reduce its active lifespan [5]. To defend against over-extraction, geothermal modeling is key. Geothermal reservoir modeling is used to monitor the health of geothermal reservoirs and predict the future state of geothermal fields by being calibrated to historical behaviour. It is important that these models are accurate because their predictions can be used to make important decisions regarding the continued operation of the wells and can act as strong evidence of profitability, driving economic investment in the field.

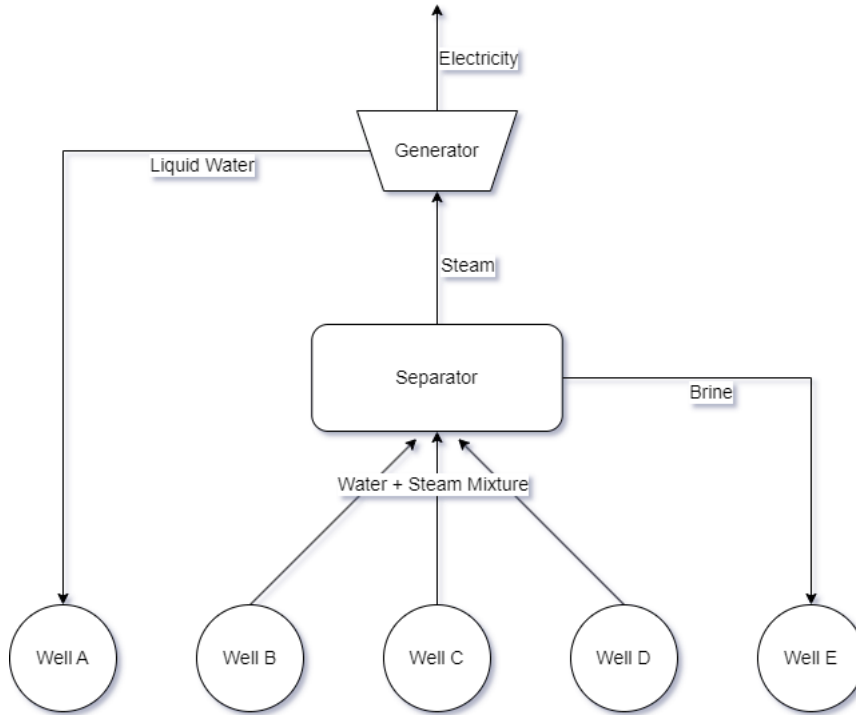


Figure 1.2: Simplified example of power generation process at a geothermal field. Hot fluid is extracted from multiple wells, pumped to a separator, and then steam continues to a turbine, where electricity is generated.

1.2 Available Data

The effectiveness of a reservoir model depends highly on the accuracy and quantity of the mass extraction data that it is provided about a single well. Dense, accurate data with a high level of certainty is required to make helpful predictions, however there is often limited data available. Often, the only frequently measured data is ‘total steam flow’ which is measured at a separator or generator, however separators are usually fed by multiple wells, so it is unclear how much mass originated from each of the wells that contributed to the separator. It is important to isolate how much mass has been extracted from each individual well. Fig. 1.2 shows that these measurements taken at the generator or separator are not representative of a single well, but rather of a number of wells, which are each contributing an unknown amount of mass to the separator.

To determine how much mass is being extracted from an individual well, as required for effective reservoir modeling, we desire measurements that were taken at the well itself. One of these tests is called a Tracer Flow Test (TFT), which can provide an estimate of the well’s mass extraction rate at that point in time. These are only taken occasionally for any given well, as the process of taking these measurements is difficult, and may require diverting flow from the separator. Because TFT measurements are taken so sparsely, often the available data for reservoir modeling is very limited. Using a combination of TFT

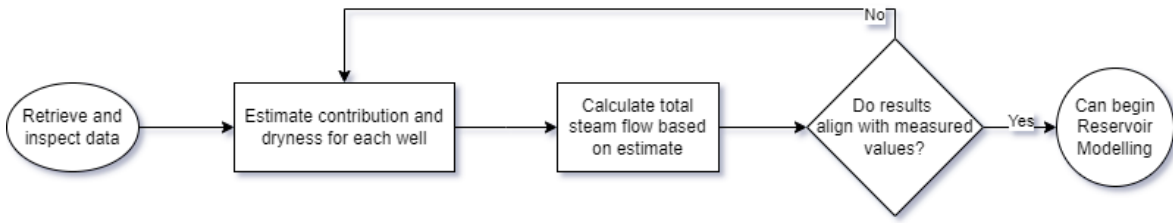


Figure 1.3: Simplified flowchart of manual process for predicting historical geothermal data.

measurements and total steam flow measurements, it is possible to predict the extraction history of a given well, however it is difficult and time-consuming to do this manually.

Manually inferring dense data from sparse measurements is an iterative process that follows the general structure of an educated ‘guess and check’. It begins by estimating the average mass contribution of each well that is connected to the separator by inspecting the relevant TFT measurements. Next, the total flow that would result from these estimates is calculated, and compared to the measured data. This allows the initial guess to be modified, aiming to change the estimate to meet the data, repeating until a subjective level of ‘closeness’ has been reached. This is not an efficient or accurate method. This method assumes that well contributions remain constant over time or follow an imposed trend, which is not a realistic assumption. Uncertainty of the produced data cannot be quantified, and the subjectivity of the process leads to added variability in results. There is a need to improve this process by automating the historical data prediction, which is the purpose of this project.

This report will begin by noting relevant research from related works, and investigating which data-driven modeling approaches could work best for this problem. Then, in Chapters 2 and 3, the iterative model development process will be explained in detail, highlighting methods that work poorly or well, before presenting the final data-driven model in Chapter 4.

1.3 Existing Approaches to the Problem

Geothermal reservoir modeling is a broad field where many approaches have been used to model geothermal wells. There are three classifications of geothermal reservoirs: high temperature above 225 degrees Celsius, low temperature below 125 degrees Celsius, and intermediate temperature between these values [6]. Higher reservoir temperatures lead to higher enthalpy of extracted fluid, which means more electrical energy can be generated. This is why it is important to use reservoir modeling to model the pressure and temperature of a geothermal reservoir over time. It gives the ability to predict the future temperature and pressure of a given reservoir, so that operational choices can be made to maximise profitability. Reservoir modeling can also ensure that we do not extract too much liquid from the reservoir to keep the reservoir stable and healthy. Reservoir modeling generally models only a single well/reservoir within a larger geothermal field, so this is why it is important to predict the data history of individual wells from total steam flow data. A number of techniques that could be used for predicting the required data will be discussed below. Their advantages and disadvantages will be discussed in order to select a preferred method to approach this problem.

1.3.1 Bayesian modeling

Bayesian modeling is a statistical method where prior distributions based on existing contextual knowledge are used to generate posterior distributions, which can be used to generate predictions, make analyses and quantify uncertainty [7]. While Bayesian modeling has been applied to a geothermal context before [8], it has not been directly applied to the data prediction scenario described above. This historical data prediction problem can be considered an allocation problem, where each well is assigned an allocation between zero

and one, and all allocations have to add to one at each timepoint. In this formulation, the most similar existing research comes from unrelated fields, such as the ‘Bellybutton Problem’, and the ‘Lions, Tigers and Bears’ problem as discussed in the book ‘Think Bayes’ [9]. This suggests that when undertaking Bayesian modeling of these situations, if it is known that all variables must sum to one, that a Dirichlet distribution is the appropriate prior. Dirichlet distributions locally act like Gaussian (normal) distributions, but also enforce that samples drawn must always sum to one.

One approach within the Bayesian framework is Monte Carlo simulation. This is a method involving repeatedly generating random samples taken from the prior distribution, and then selecting the samples that go ‘near’ data points as feasible solutions, which are then used to generate the posterior distribution [10]. This can help us to quantify the uncertainty [10] of model estimates through results such as prediction intervals. However, this can lead to overfitting of uncertain and potentially noisy data. To prevent overfitting, ensemble methods can be used. These are methods such as Bayesian Voting, that have many models run simultaneously and then take a weighted average of all their results to give a final posterior [11]. This method of combining multiple ‘weak’ regressors into one ensemble model helps to reduce overfitting, because each individual model is trained on only a subset of the training data. This means they are less fit (and may even underfit) as individuals, but as a whole come together to give good results.

When engaging in traditional Bayesian modeling and computation, it is important to know the relevant likelihood function or at least be able to estimate it by computing it numerically. In some situations this is not available and this is when Approximate Bayesian Computation (ABC) can be used to sidestep the likelihood function [12, 13]. ABC is a likelihood free inference method where samples are generated from the prior distribution, and run through the model to examine their similarity to the posterior distribution using summary statistics such as the mean, variance, range and maximum [14]. Using a metric such as Euclidean distance, these statistics are compared to those of the posterior, and if they are ‘close’ enough, that sample is accepted. As this is an ensemble method [13], this is then repeated to generate many samples, making it resistant to overfitting. The choice of summary statistics is important and often requires high contextual knowledge to choose appropriate statistics [15], however the process can be more computationally efficient than other Bayesian methods due to the summary statistics reducing the dimensionality of the model. Without expert domain knowledge, it would be difficult to apply to this problem, because neither the appropriate prior distribution nor summary statistics are known in this context.

Gaussian Process Regression (GPR) is a Bayesian machine learning method that involves using a kernel to define a covariance matrix between unknown random variables. GPR can only be used when certain assumptions are met, including that the dataset, and the associated noise within, follow Gaussian distributions [16]. This dataset cannot have perfectly Gaussian distributions, as the physical requirement that all mass flows at a given time sum to the total mass flow ensures that samples are not independent. Also the mass flows must remain between a lower bound of zero and an upper bound of the total mass flow, so a true Gaussian distribution is not appropriate, because a Gaussian distribution is unbounded. However, it may be possible to assume that the noise is ‘close enough’ to Gaussian if a Dirichlet distribution (as mentioned above) is used, because this acts locally as a Gaussian Distribution, so it is worth investigating whether this method can be applied to the problem.

1.3.2 Non-Bayesian Machine Learning Approaches

Following the lead of the oil and gas industries, the geothermal industry’s use of machine learning has been increasingly exponentially since 2018 [17]. In a geothermal context it has previously been used in the context of well exploration, heat flux prediction, reservoir modeling, and above-ground component modeling, but has not been used previously to estimate historical geothermal mass extraction data [18], as in this situation. Different machine learning approaches are discussed below.

Machine learning (ML) is the process of creating algorithms that learn how to improve themselves from their experiences [19]. In practice, a machine learning model is a model that will get better with the more data you train it on. There are many types of learning algorithms. Some of these will be discussed below. The goal of a machine learning algorithm is to find a model that minimises a chosen loss function [20]. When each additional piece of training data is fed into the algorithm, it will work out how to use the additional information to keep minimising the loss function, getting better and better until it runs out of training data. However, this process can lead to overfitting, which is when a model learns too effectively on the training data to the point that it captures the noise within the training data, instead of capturing the true trend as desired.

To avoid overfitting, a regularisation term is often added onto the loss function. Regularisation can come in many different forms, such as LASSO regression, ridge regression, Tikhonov regularisation [21], singular value decomposition, early stopping criteria, and more. LASSO regression aims to decrease the number of non-zero parameters in the model and performs feature shrinkage, while ridge regression decreases the magnitude of parameters in the model [22], and Tikhonov regularisation aims to ‘smooth’ out a model by minimising its first, second or higher order derivative(s) [21]. The regularisation term that should be used is determined by contextual knowledge regarding the expected behavior of the system. If it is expected that the output should be smooth, Tikhonov regularisation may be appropriate, while if sharp changes are expected then LASSO regression may be preferred.

There are many types of existing machine learning algorithms. Random forest regression use an ensemble of decision tree regressors to predict the desired output [23]. The decision trees each involve a sequence of splits from the root node down to the leaf nodes, where the leaf node will assign the input data a predicted output. Each of these decision tree regressors can be given a maximum depth and complexity, and the large number of trees effectively being ‘averaged’ make it very difficult to overfit a random forest model [23]. Random forests also require relatively few computations, and are therefore an appealing machine learning model. Random Forests are especially effective in classification problems, but this problem is a regression problem, so it may not be as well suited in this context.

Neural Networks are a machine learning method which involves creating a sequence of many layers of neurons, which apply different weightings to data. These models need to be provided large amounts of training data [24] to become effective predictive models because there are so many weightings which need to be determined by the algorithm. Deep learning methods such as neural networks often require large amounts of data to become more effective than simpler models [25]. The amount of data that a neural network could be provided is limited in this context by the data which has been measured at the geothermal site. Keeping this in mind, a traditional neural network may not be ideal in this context. However, it may be possible to sidestep this issue by generating additional simulated training (and testing) data to feed the models. There are existing geothermal simulations that could generate the data required. Data augmentation could also be used to generate additional pieces of data [24] by intentionally modifying existing datasets.

1.3.3 Physics Informed Machine Learning

The issue with the above machine learning methods is that the outputs for each well are not guaranteed to sum to the known total mass flow. This means that while predictions may look accurate, they will likely be physically impossible. To fix this issue, physics-informed constraints can be added to the model, to ensure that physical laws are being obeyed. For example, this can ensure that all contributions sum to the total extraction rate and that each well’s mass flow rate will change smoothly over time. One way to do this is by using a Physics-Informed Neural Network (PINN) [26]. PINNs are able to incorporate these physical laws into their loss functions, where multiple functions can be minimised at once. The final loss function can become a sum of many individual loss functions relating to different physical laws, which are summed together with chosen weights which relate to the relative importance of those laws. Upon iteration through the network, the loss function will be minimised, thus specifying that the physical laws are being approximately

followed as they should. There are also loss functions applied for the accuracy of the data, and for the initial conditions and boundary conditions being met. PINNs generally work in discretised parameter spaces, which is appropriate to use in this context, because the data taken has been taken at discrete moments in time. Bayesian PINNs (B-PINNs) can even be used to combine the effects of Bayesian methods with Physics Informed Neural Networks [27].

Another benefit of using physics informed machine learning is that it can model any physical scenario, as long there is a differential equation which models the physical phenomenon in question. There is no rule on how ‘complicated’ the equation can be and it is not required to have an analytical solution to be applied to a PINN. As geothermal physics can be represented by a highly complicated Partial Differential Equations (PDEs), this is ideal. It has also been used to model solutions in the field of fluid mechanics, and used on the notoriously difficult Navier-Stokes equations [28]. ML-augmented simulation has also been used by ANSYS, a premier fluid simulation tool, to "speed up simulation by factors of 100x", according to an ANSYS blog [29].

1.3.4 Solving as an Optimisation Problem

The key idea behind physics informed machine learning is that of the loss function [20]. Another approach that could be taken is to treat the loss function as the objective of a traditional minimisation problem, and to solve this using other optimisation techniques. A loss function is a measure of the error between a predicted solution and a set of measured data [20, 30]. If a differentiable loss function can be defined for the geothermal system then the gradient of the loss function can be set to zero to find the analytical minima of the loss function. The associated decision variable values will be the optimal values. This method, referred to as Loss Function Optimisation (LFO) in this report, appears to be a new technique which has not been used previously. Because this is not a traditional machine learning technique involving training and testing data, this method does not require excess data, and modifications to the loss function can easily be made in order to add regularisation to the model.

Inverse problems are a class of ill-posed problem where the goal is not to predict the output data, but the input parameters [31]. It is another way of considering this problem because there are more parameters than TFT data points in this context, meaning it is ill-posed. There are many techniques for solving inverse problems, especially if they are linear inverse problems. These can be solved analytically using linear algebra to produce optimal parameter estimation. It is a reliable and fast method, so could be used effectively in this scenario. To apply it, a smaller number of parameters will need to be selected, such as coefficients of a polynomial, rather than allowing the contribution of each well at each timepoint to be an individual parameter, as this would lead to the problem becoming too large to solve efficiently.

1.3.5 Uncertainty Quantification

Uncertainty quantification is another goal of this project. The output of the models should be able to have quantifiable uncertainty, so that this can be provided to the predictive geothermal models along with the mean predictions generated. This can, in turn, allow the geothermal modelers to produce an uncertainty estimate along with their outputs, which can help decision-makers to make the best possible decisions for the future of their geothermal fields. There are a number of techniques that can allow uncertainty to be quantified. Bayesian methods do not produce a single estimate, but instead a distribution of estimates, called the posterior. This in-built method of uncertainty quantification is a benefit of using Bayesian methods [7] such as Bayesian regression and GPR.

Monte Carlo simulation for uncertainty propagation is another simple way to achieve uncertainty quantification by simulating uncertainty in the input data. There are more complicated methods of simulation such as Markov Chain Monte Carlo (MCMC) methods that expand on this method [32], but MCMC is more suited to a Bayesian context [33].

The benefit of Monte Carlo simulation is that it can be applied to conduct uncertainty quantification even for non-Bayesian methods such as optimisation approaches. Monte Carlo simulation is computationally intensive because it requires generating many samples, and running each of these through the model [34]. This process can be sped up through parallelisation as Monte Carlo simulation can be easily run in parallel, due to samples being independently generated [35]. For efficient models it is often the best choice due to its simplicity and because it enables the construction of a posterior distribution. Statistics such as the standard deviation and variance can then be calculated from this posterior distribution, thus quantifying the uncertainty of predictions.

1.4 Research Approach

The scope of this research involves building a computational model to effectively infer the data history of a set of wells in a geothermal field. This data should be suitable to provide to a predictive geothermal model, and should be accompanied by the appropriately quantified uncertainty of the produced data. The reservoir models using the historical data are out of scope for this project, as it focuses only on historical data prediction. The physical scope pertains to a subsection of a geothermal field and its associated wells (real or simulated), which will provide a realistic amount of sparse data to the model. Throughout this report, it will be assumed that all wells feed into a single separator (as shown in Fig. 1.2). It would be possible in the future to expand the scope of the model to include more complex geothermal systems which have components interlinked in more complicated ways.

Myself and my project partner Michael Terekhin will each attempt to build a model that solves this problem. Personally, I will attempt to use optimisation methods to find the theoretically optimal solution for this historical data prediction. This will begin with a naive model which assumes a constant extraction from each well and over time relax this assumption to build a model that has more flexibility and can better represent the true behaviour. Then, other forms of complexity will be introduced, such as Tikhonov regularisation and parameter tuning, to improve the output of the model on a simulated dataset. Lastly, the model's uncertainty will be quantified to ensure that reservoir modelers using the historical data generated can understand the uncertainty in these predictions, so that they can correctly propagate the uncertainty of their own predictions.

Concurrently, my project partner Michael Terekhin will aim to solve the same problem using Gaussian Process Regression [36]. He will also start with a simpler model, and slowly build its complexity until it can account for the processes occurring in a given geothermal system. Once Michael and I have both created our models, the results of our respective models will be compared to see which of the models is more effective under different situations. Chapters 2 and 3 explore how my models were created through an iterative process of starting with a simplified version of the situation, and then adding complexity to get a better model over time.

Chapter 2

Loss Function Optimisation (LFO)

This chapter will outline the iterative LFO model development process. Firstly, the tools used will be outlined. Next, an initial Loss Function Optimisation (LFO) model will be formulated, before iteratively relaxing constraints of this initial model to improve its capability to match the true data. Throughout this chapter and chapter 3, simulated data will be used for model testing and development. This dataset contains five wells and is discretized into 41 timepoints, each of which represents one month. A true data history does exist for this dataset, so the model predictions can be compared with this to evaluate the accuracy of the model.

2.1 Tools Used for Model Development

A number of Python packages have been used to develop the models explained in this section. SymPy is the key package used as it allows for symbolic equation manipulation, including matrix multiplication, differentiation and substitution of variables, which is used throughout this Chapter. NumPy, SciPy, Matplotlib, Pandas and Time packages are also used. Furthermore, Jupyter notebook will be used as the main environment to develop and tune the model. Jupyter notebook is a web tool for using interactive code documents. Code documents can include Python code, equations, plots, Markdown text and more. It is preferable to a traditional coding environment such as Visual Studio Code or PyCharm because it has the ability to run only certain code chunks at a time, as opposed to traditional programming scripts which run the entire program. This reduces the computation cost of developing the model, and also makes it simpler to visualise the model output in the form of plots nested in between code chunks.

Australian Research Data Commons (ARDC) Nectar Research Cloud is an Australian cloud computing service specifically designed to enable large-scale computing research for academics across Australia and New Zealand [37]. It has a Jupyter Notebook service that allows Notebooks to be run remotely [38], which was used to run the majority of code during this project. The existing scientific Python environment will be used for this project, as this included Sympy, Matplotlib, Numpy, and all other packages that were required.

2.2 Formulating a Constant LFO Model

Initially, a simplified version of the problem will be formulated where all wells have a constant average extraction rate. This assumption can be written as

$$W_i(t) = w_i, \quad (2.2.1)$$

where $W_i(t)$ is the generalised expression of the (possibly time-varying) extraction rate of well i , while w_i is the constant extraction rate of well i . This assumption will simplify the formulation of an initial loss function. A loss function is a function which takes in a set of

decision variables, and evaluates how well these map to data by evaluating the associated ‘error’ or ‘loss’. In this case, there are two losses which will contribute to the loss function. Firstly,

$$\mathcal{L}_1 = \sum_{t=1}^T (M_t - \sum_{i=1}^n z_{i,t} w_i)^2 \quad (2.2.2)$$

shows the total squared loss of the estimated total mass flow against the true total mass flow, where $\mathcal{L}_1$ represents the total squared loss against these bulk measurements, n is the amount of wells in the geothermal system, T is the amount of discretized timepoints, M_t represents the true total mass flow measured at time t , and $z_{i,t}$ represents whether Well i was on or off at time t (where zero represents off and one represents on). Secondly,

$$\mathcal{L}_2 = \sum_{i=1}^n \sum_{t=1}^T d_{i,t} (m_{i,t} - w_i)^2 \quad (2.2.3)$$

shows the squared error of the predicted mass flow rates against the known TFT values, where $\mathcal{L}_2$ represents the total loss against these TFT measurements, $m_{i,t}$ represents the measured TFT mass flow of Well i at time t , and $d_{i,t}$ represents whether a TFT measurement was or wasn’t taken for Well i at time t (where zero represents unmeasured and one represents measured). Eq. (2.2.2) and Eq. (2.2.3) can then be weighted with a regularisation parameter and summed together to give the equation of the total initial loss function,

$$\mathcal{L}_c(w_1, w_2 \dots w_n) = \sum_{t=1}^T (M_t - \sum_{i=1}^n z_{i,t} w_i)^2 + \lambda_1 \sum_{i=1}^n \sum_{t=1}^T d_{i,t} (m_{i,t} - w_i)^2, \quad (2.2.4)$$

where $\mathcal{L}_c(w_1, w_2 \dots w_n)$ represents the total loss from both sources and λ_1 is a regularisation parameter that controls the relative weighting given to the TFT losses relative to the bulk measurement losses. This is the initial loss function that will be used to conduct Loss Function Optimisation (LFO).

2.3 Solving LFO Analytically

The purpose of forming the loss function in Eq. (2.2.4) is to find its minimum. Minimising the loss function is equivalent to finding the solution that fits the data best. To solve LFO, we need to solve

$$\hat{w} = \arg \min_w \left(\mathcal{L}_c(w_1, w_2 \dots w_n) \right). \quad (2.3.1)$$

Initially, this was solved by using a built-in SciPy minimisation solver, which took a long time to converge. To make the solving process more efficient, a new method of solving Eq. (2.3.1) was devised. By differentiating the loss function against each decision variable w_i and setting each of the resulting equations to 0 simultaneously, the w_i values that give the minimum of the loss function can be calculated analytically according to

$$\frac{\partial \mathcal{L}_c(w_1, w_2 \dots w_n)}{\partial w_i} = 0 \quad \forall i \in [1, n]. \quad (2.3.2)$$

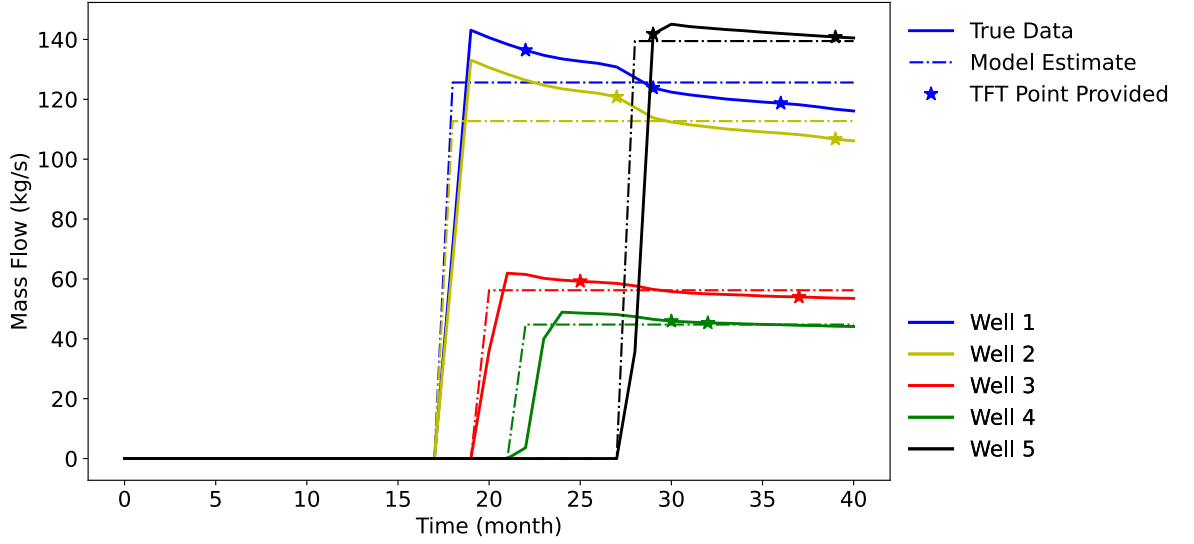
Because the loss function deals with squared errors, these derivatives are at most linear in each w_i . These n linear equations can then be composed into a matrix equation,

$$L\hat{w} = b, \quad (2.3.3)$$

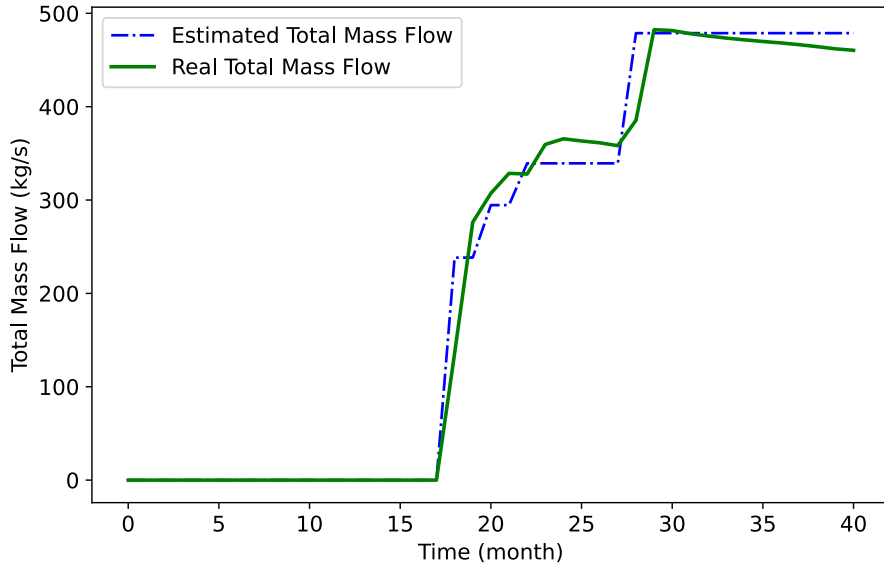
where $\hat{w}$ represents the vector of optimal w_i values, L contains the non-constant coefficients in Eq. (2.3.2) such that $L_{i,j}$ = the j ’th coefficient of $\frac{\partial \mathcal{L}(w_1, w_2 \dots w_n)}{\partial w_i}$ and b_i represents the negative of the constant term of $\frac{\partial \mathcal{L}(w_1, w_2 \dots w_n)}{\partial w_i}$. The optimal $\hat{w}$ vector can then be found by

evaluating this expression, by calculating $L^{-1}b$ or other equivalent technique.

Fig. ?? shows the results of solving this initial model. The results produced are a realistic simplification of the real behaviour of the system, based on the assumption of a constant extraction rate. The predicted values for w_i are all close to the true average extraction rate of each well, and are also close to the TFT values provided. Naturally, because more than one TFT point is being provided for each well, it is impossible to match these exactly while assuming that the extraction from each well is constant. RMSE for this estimate against the true mass flow values is 10.36 kg/s. The RMSE will be used to quantify the effectiveness of models throughout this report, and a table summarising these RMSEs is shown in chapter 5. Fig. ?? shows the estimated total mass flow over time against the



(A) Well-specific estimated and real mass flows



(B) Total estimated and real mass flows

true value. This shows a reasonable match, as the loss function also seeks to minimise the error against this known total mass flow value, however this total is still limited to being a piecewise constant graph, so it does not capture the finer behaviour of the system. For future models, the total mass flow plots will not be shown in the body of the report, but can be found in Appendix A.

2.4 Calculating a Balanced Regularisation Parameter

In Eq. (2.2.4) the parameter λ_1 represents a weighting of the importance given to the sum squared error against the TFT measurements relative to the importance given to the sum squared error against the bulk measurements. To find a balanced value for this, we can find λ_b which predicts an equal error from $\mathcal{L}_1$ and $\lambda_b \mathcal{L}_2$, on average. It is assumed that the error within the bulk term will be approximately n times that of the error within an individual well measurements, because bulk values are n times larger, as they are made from the contributions of n wells. By substituting into Eq. (??), it can be shown that the average loss $\mathcal{L}_{ave}$ can be broken down into its two contributing sources (TFT points and bulk values) according to

$$\mathcal{L}_{ave} \approx \lambda_1 D E_{ave}^2 + T n^2 E_{ave}^2, \quad (2.4.1)$$

where E_{ave} is the average error of a single well prediction at a given time, D is the total number of TFT points provided, T is the total number of timepoints, and n is the total number of wells. This can be rearranged to express the balanced value of λ_b as

$$\lambda_b = \frac{T n^2}{D}. \quad (2.4.2)$$

For the simulated dataset, when 11 TFT points were provided (as used for model development), this produced a λ_b value of 93.18. $\lambda_1 = \lambda_b = 93.18$ will be used throughout model development. This balanced lambda value depends on the proportion of points which are provided as TFT points, so if more TFT points are provided, λ_b will decrease to compensate.

2.5 Relaxing LFO Assumptions

Eq. (2.2.4) is the initial loss function, however it does have some shortcomings, because it is designed whilst making a number of key assumptions. Firstly, it assumes that the contribution from each well is constant over time. This is an oversimplification of real well behaviour, and is a poor assumption, but it helps us to greatly simplify the expression to create an initial loss function. This will be relaxed by introducing linear time-dependence into the system. Instead of setting $W_i(t) = w_i$, it is now defined as

$$W_i(t) = w_{c,i} + w_{g,i}t, \quad (2.5.1)$$

where $w_{c,i}$ represents the y-intercept of the line representing the extraction rate over time of well i , and $w_{g,i}$ represents its gradient. The new loss function is

$$\mathcal{L}_l(w_1, w_2 \dots w_n) = \sum_{t=1}^T \left(M_t - \sum_{i=1}^n z_{i,t} (w_{c,i} + w_{g,i}t) \right)^2 + \lambda \sum_{i=1}^n \sum_{t=1}^T d_{i,t} \left(m_{i,t} - (w_{c,i} + w_{g,i}t) \right)^2. \quad (2.5.2)$$

This results in a similar matrix equation as in Eq. (2.3.3), however the w vector will now be twice as long because there are now two coefficients (w_c and w_g) for each well, and therefore $A \in \mathbb{R}^{2n \times 2n}$. This can again be differentiated symbolically and solved to get the following results shown in Fig. 2.1. These results look similar to the results of the initial model, however because of the linear time-dependence, they are able to more effectively match the given data, and this reduces the RMSE to 10.34.

It appears visually that the RMSE should decrease far more than this small decrease of 0.02. This happens because RMSE is dominated by larger errors, due to the squaring of errors that occurs. This means that this large visual improvement does not have as much of an effect on the RMSE score because that is dominated by the few large errors in the prediction. These occur as each of the wells are turned on, when wells are on for only part of a month. This can be accounted for by redefining the z matrix. Previously, the z matrix was a binary matrix, where $z_{i,t}$ is one if well i was on during month t , and zero otherwise. However, because monthly time-discretization is very coarse, often a well

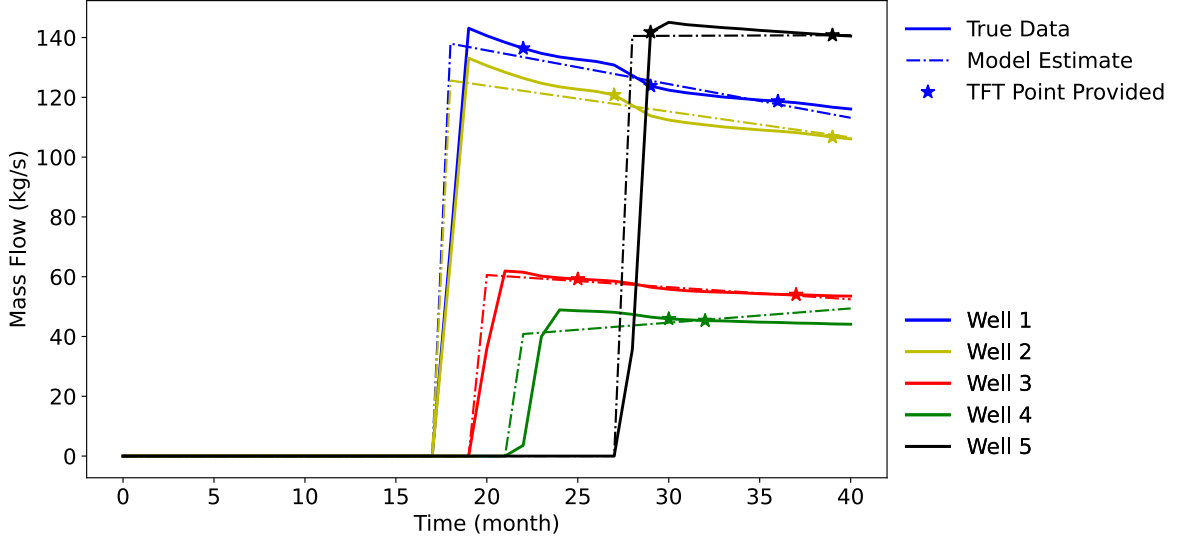


Figure 2.1: Mass flows predicted by linear LFO against true mass flows. This improves on the constant LFO model by allowing the model to represent the real behaviour more accurately, and decreases the RMSE to 10.34.

may be turned on halfway through a month. This would strongly effect the mass-flow measurement for that month, leading to an abnormally high effect on the RMSE. This can be seen in Fig. 2.1 where well 4 turns on partway through month 22. During this month, the true mass flow line increases to about 10% of the value it would be if it had been turned on throughout the whole month. However, the predicted mass flow spikes up to the full expected value because the model cannot yet handle the case of a well turning on for only part of a month. This causes the prediction to be artificially ‘pulled down’ by this early datapoint, resulting in an increasing trend for Well 4 when one should not exist.

This can be fixed by allowing the z matrix to contain non-binary values, which represent the amount of the month that the well was turned on for. For example, $z_{i,t} = 0.2$ if well i was on for 20% of month t . All $z_{i,t}$ values will be between 0 and 1 (inclusive). Fig. 2.2 shows how this improves the results for the same linear LFO model as in Fig. 2.1. The predictions are no longer being ‘pulled down’ by an early low measurement from a partially-on month, so they exhibit behaviour more reflective of the true well behaviour. For example, Well 4’s prediction is now much flatter, which represents the true data more accurately. The RMSE decreases from 10.34 to 1.22, showing the large improvement that is made by this small change.

The logical next step is to continue relaxing the assumption of linear time-dependence by creating a quadratic model, cubic, quartic, and so on. To simplify this process, a model is created that can define each well’s extraction rate over time as an arbitrarily high order polynomial (AHOP). A new parameter k is introduced that represents the highest degree of the polynomial. The arbitrarily high order polynomial is defined as

$$W_i(t) = \sum_{j=1}^{k+1} w_{i,j} t^{j-1}, \quad (2.5.3)$$

where w is now an $n \times k$ matrix. The associated loss function can now be defined as

$$\mathcal{L} = \sum_{t=1}^T \left(M_t - \sum_{i=1}^n z_{i,t} W_i(t) \right) + \lambda \sum_{i=1}^n \sum_{t=1}^T d_{i,t} \left(m_{i,t} - W_i(t) \right)^2. \quad (2.5.4)$$

Results for AHOP LFO with $k = 3$ (cubic) are shown in Fig. 2.3. These are worse

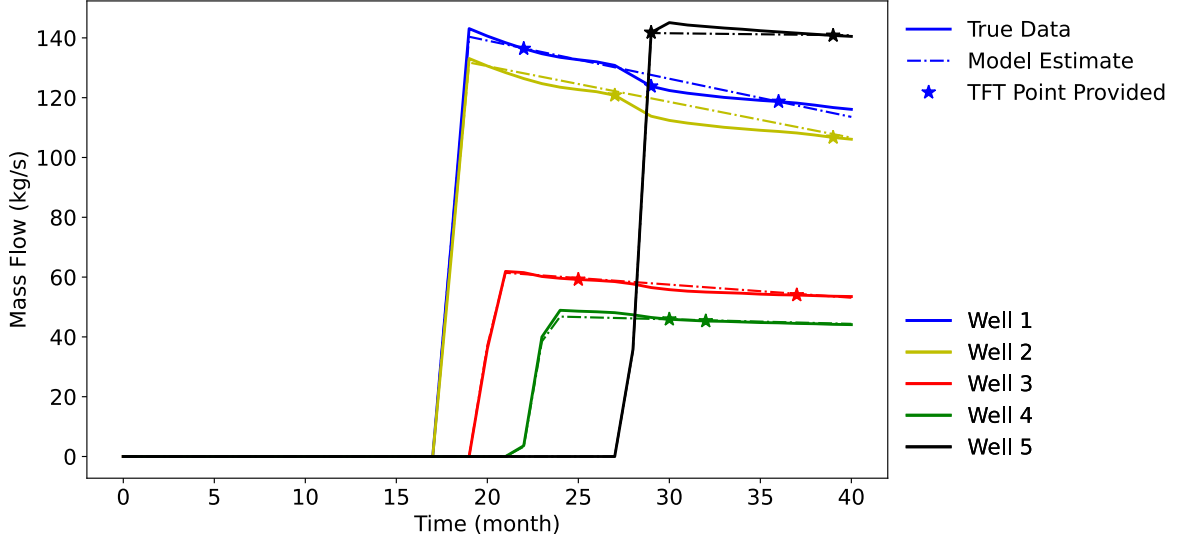


Figure 2.2: Mass flows predicted by linear LFO using a non-binary z matrix. This allows the linear model to only go up part of the way if a well turns on in the middle of a month, allowing it to match the data better. The RMSE of this prediction is 1.22, which is a significant decrease.

predictions than those produced by the constant and linear models, because the cubic model has more decision variables and is overfitting to the provided data. It is able to ensure predictions meet the TFT points even more accurately than the other models, by introducing wild variation that does not exist in reality. Figure A.2 shows how the total mass flow data is also perfectly matched. Due to this overfitting, the RMSE increases to 13.24 for this prediction, but it is important to enable the model to exhibit higher-order behaviour, because real geothermal wells will exhibit more complex behaviour. The AHOP model produces even more rapidly varying and inaccurate predictions for higher values of k . To remedy this, a regularisation term(s) will be added to Eq. (2.5.4) with the aim of smoothing or ‘flattening’ the results.

2.6 Applying Regularisation to LFO

There is a number of ways Tikhonov regularisation could be implemented to improve my model fit to this data. The first way attempted was conducting ridge regression by adding an L_2^2 norm term to Eq. (2.5.4), according to

$$\mathcal{L}_{Ridge} = \mathcal{L} + \lambda_2 \sum_{i=1}^n \sum_{j=1}^{k+1} w_{i,j}. \quad (2.6.1)$$

This did reduce the size of the coefficients as expected, but because the sum squared error is being used, and the lower order polynomial coefficients tend to have a much larger magnitude, it much more heavily penalised the lower order coefficients (such as the constant and linear terms) than the higher order terms (such as the quadratic and cubic coefficients). This did smooth out the results as shown in Fig. (2.4) (A) and reduced the RMSE to 3.79. However, it also artificially forced the y-intercept of these polynomials towards 0, which is not a realistic constraint to add to the model, and biases the solution. For this reason, the ridge regression term was removed from the model, and replaced with a new form of Tikhonov regularisation.

To ensure the regularisation term ‘flattened’ the results without artificially decreasing the constant term towards zero, Tikhonov regularisation was conducted on the first and second time-derivatives of $W_i(t)$. Two new terms are added to the loss function, giving a

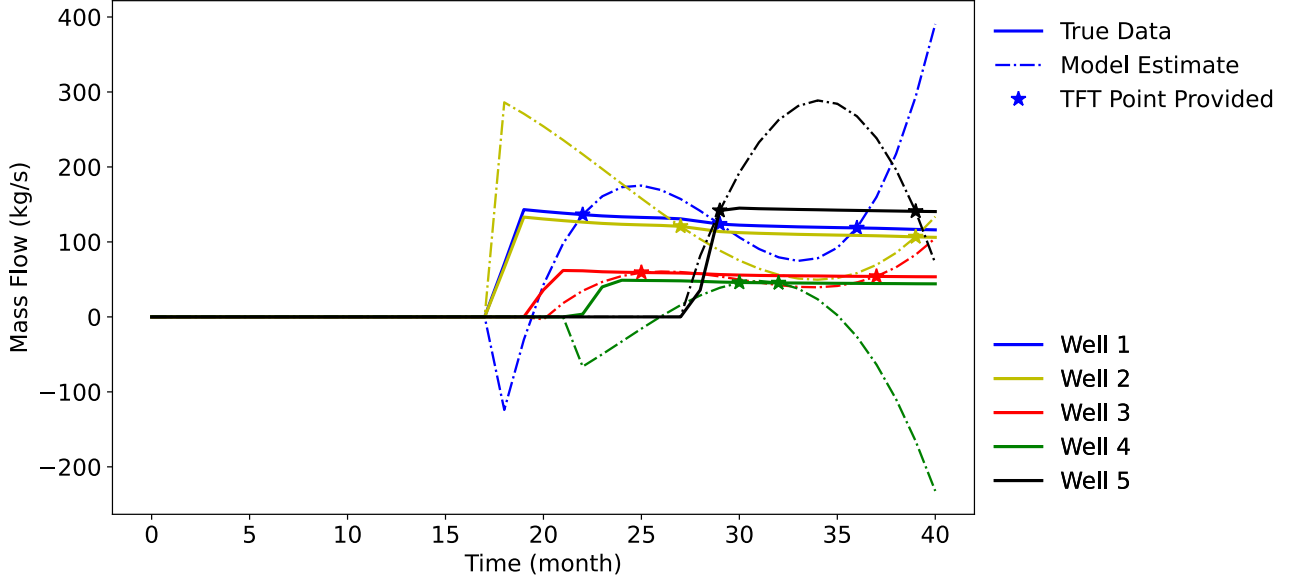


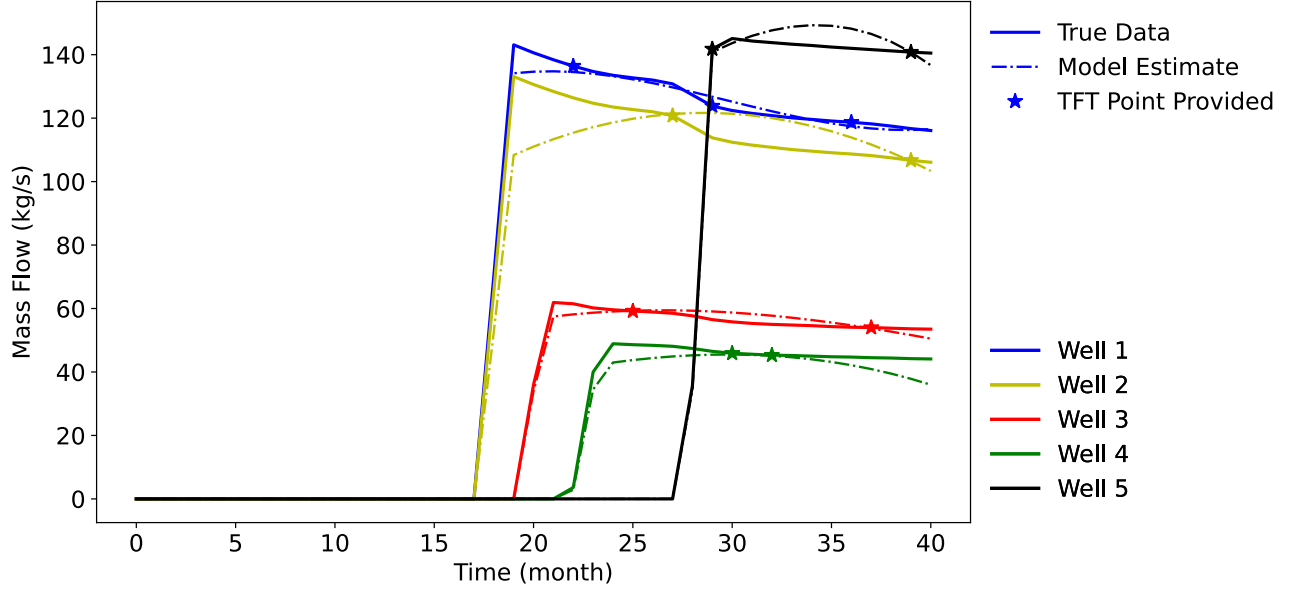
Figure 2.3: Mass flows predicted by AHOP LFO for $k=3$ (cubic) against true mass flows. This prediction is clearly a poor one, as it is overfitting to the provided data. It is nearly perfectly matching each TFT point, but is modelling unrealistic behaviour between these points. The RMSE of this model is 13.24, which is significantly worse than previous models.

final loss function of

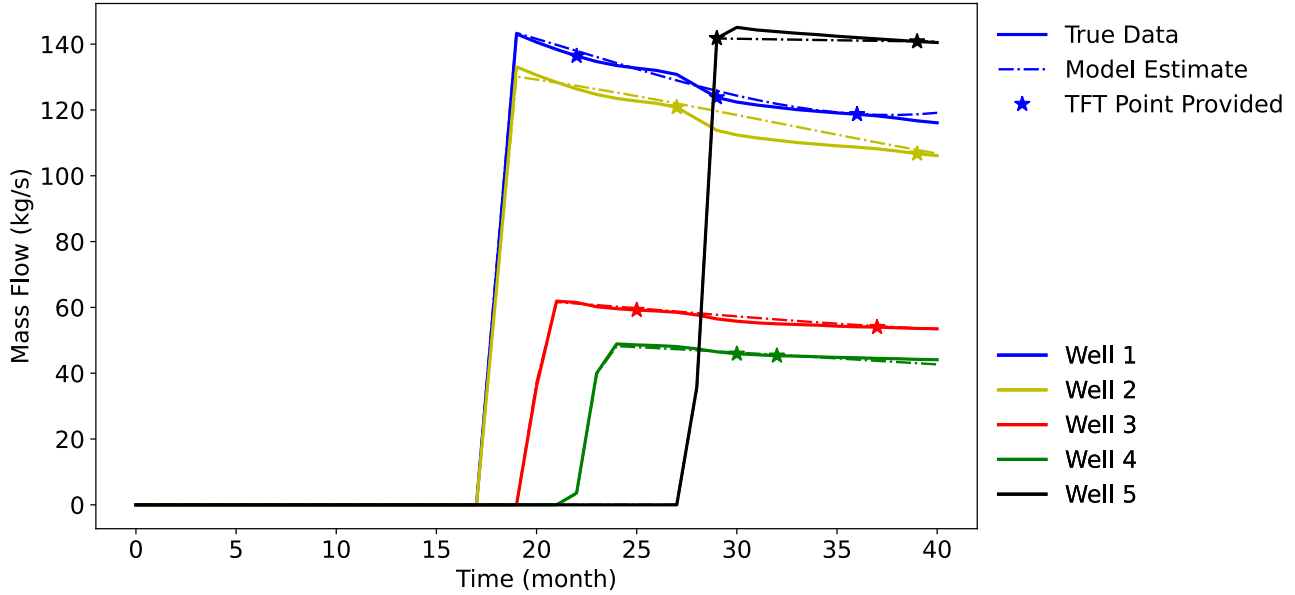
$$\begin{aligned}
 \mathcal{L}_{Tik} = & \sum_{t=1}^T (M_t - \sum_{i=1}^n z_{i,t} W_i(t) + \lambda_1 \sum_{i=1}^n \sum_{t=1}^T d_{i,t} (m_{i,t} - W_i(t))^2 \\
 & + \lambda_2 \sum_{i=1}^n \sum_{j=1}^T \left(\frac{dW_i(t)}{dt} \Big|_{t=j} \right)^2 + \lambda_3 \sum_{i=1}^n \sum_{j=1}^T \left(\frac{d^2 W_i(t)}{dt^2} \Big|_{t=j} \right)^2.
 \end{aligned} \tag{2.6.2}$$

The prediction using this loss function is shown in Fig. 2.4 (B). The RMSE is reduced from 13.24 to 1.33. This is still not as low as the RMSE of the linear model using non-binary z , indicating that the regularisation parameters used for this prediction were sub-optimal.

The next logical step was to conduct a grid or metaheuristic search across the parameter space of λ_1 , λ_2 and λ_3 . Unfortunately the available computation power proved too low to find effective results using LFO. Instead, the problem was re-formulated as a linear inverse problem, where the same loss function as in Eq. (2.6.2) is expressed as an ill-posed parameter estimation problem. This can be solved much more efficiently, and will allow us to optimise to a higher degree. Chapter 3 explores this process in detail.



(A) Ridge-regularised mass flow predictions



(B) Tikhonov-regularised mass flow predictions

Figure 2.4: Mass flows predicted by AHOP LFO for $k=3$ (cubic) under different regularisation schemes, against true mass flows. Fig. (a) shows the results produced by ridge regularisation, which ‘flattens’ the results significantly, but biases results by artificially forcing the y-intercept of each well’s function to be zero. The RMSE of this prediction is 3.79. Fig. (b) shows the Tikhonov-regularised results, which successfully ‘flatten’ the prediction, reducing overfitting, but still allow the model enough freedom to meet the results with high accuracy. The RMSE for this prediction is 1.33.

Chapter 3

Linear Least Squares Optimisation (LLSO)

This chapter will discuss how the LFO model can be reformulated as an inverse problem, giving rise to a Linear Least Squares Optimisation (LLSO) model. LLSO is significantly more efficient, and the speedup between the two models is discussed below. Lastly, a particle swarm optimisation (PSO) metaheuristic will be used to find optimal regularisation parameter values to produce an optimised final model.

3.1 Converting LFO to an Inverse Problem

The general form of a linear inverse problem is

$$y = Ax + e \quad (3.1.1)$$

where y represents the outputs of the forward model A , x represents the parameters of interest in the input space and e is the unknown noise. The forward problem is well-posed, meaning it has unique solutions, and is stable under small perturbations of the parameters x . Trying to predict y in this scenario is a standard forward problem, while trying to determine x is an ill-posed inverse problem. For this specific inverse problem, A is decomposed according to $A = A_1 A_2$. Eq. (3.1.1) can now be rewritten as

$$y \approx A_1 A_2 x, \quad (3.1.2)$$

where A_1 contains measured data such as when the wells were on/off and when TFT measurements were provided. A_2 is a block diagonal matrix where each block contains the forward polynomial model. Note that Eq. (3.1.2) does not include an error term e , and changes $=$ to $\approx$ to compensate. This notation will be used henceforth. The vector of all well predictions at all timepoints q can also be defined by

$$q = A_2 x. \quad (3.1.3)$$

This is the desired quantity of interest, as it contains the predicted mass flow for each well at each discretised point in the time-domain. This is an important quantity to define because to enforce smoothness on the resulting polynomial, Tikhonov regularisation will be applied to this vector q . Eq. (3.1.4) is an example case of Eq. (3.1.2) which shows how the existing problem data labelled M_t , $m_{i,t}$, $z_{i,t}$ and $d_{i,t}$ (As defined in Eq. (2.2.2) and Eq. (2.2.3)) can be input into the LLSO model:

$$\begin{bmatrix} m_{11} \\ m_{12} \\ \vdots \\ m_{1T} \\ m_{21} \\ \vdots \\ m_{nT} \\ M_0 \\ M_1 \\ M_2 \\ \vdots \\ M_T \end{bmatrix} \approx \begin{bmatrix} d_{11} & 0 & \dots & 0 & 0 & \dots & \dots & 0 \\ 0 & d_{12} & & \vdots & \vdots & & & \vdots \\ \vdots & & \ddots & \vdots & \vdots & & & \vdots \\ 0 & \dots & \dots & d_{1T} & \vdots & & & \vdots \\ 0 & \dots & \dots & \dots & d_{21} & & & \vdots \\ \vdots & \vdots & & & & \ddots & & \vdots \\ \vdots & \vdots & & & & & \ddots & \vdots \\ 0 & \dots & \dots & \dots & \dots & \dots & \dots & d_{nT} \\ z_{11} & 0 & \dots & 0 & z_{21} & 0 & \dots & 0 \\ 0 & z_{12} & & \vdots & 0 & z_{22} & & \vdots \\ \vdots & & \ddots & \vdots & \vdots & & \ddots & \vdots \\ 0 & \dots & \dots & z_{1T} & 0 & \dots & \dots & z_{nT} \end{bmatrix} \begin{bmatrix} 1 & t_1 & t_1^2 & \dots & t_1^k & 0 & \dots & \dots & \dots & 0 \\ 1 & t_2 & t_2^2 & \dots & t_2^k & \vdots & & & & \vdots \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & & & & \vdots \\ 1 & t_T & t_T^2 & \dots & t_T^k & 0 & \dots & \dots & \dots & 0 \\ 0 & \dots & \dots & \dots & 0 & 1 & t_1 & t_1^2 & \dots & t_1^k \\ \vdots & & & & \vdots & 1 & t_2 & t_2^2 & \dots & t_2^k \\ \vdots & & & & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & \dots & 0 & 1 & t_T & t_T^2 & \dots & t_T^k \end{bmatrix} \begin{bmatrix} x_{10} \\ x_{11} \\ x_{12} \\ \vdots \\ x_{1k} \\ x_{20} \\ x_{21} \\ x_{22} \\ \vdots \\ x_{nk} \end{bmatrix} \quad (3.1.4)$$

Upon running both the LFO and LLSO models with a balanced parameter λ_b and $k = 2$, similar results are produced by both methods. The RMSE between the LFO and LLSO predictions was only 0.697 kg/s, which shows the LLSO model is functioning as expected.

3.2 Speed Comparison of LFO and LLSO

LLSO is a much more efficient model than LFO. Speedup is defined as the percentage decrease of time taken to solve the LLSO model, relative to the LFO model. This is evaluated this for many subsets of the simulated data, to determine whether the speedup is relative to the size of the problem. T , the number of timepoints, is varied between six and 43, while k , the highest degree of the polynomial, is varied between zero and five to test if the speedup is dependent on these values. Results are shown in Table (3.1).

Table 3.1: Speedup of LLSO against LFO for different T and k values, for $n=5$, rounded to nearest 100%. This shows much higher speedup for larger problems.

Speedup [%]	$k = 0$	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$
$T = 6$	1200	2400	6500	9700	17000	10400
$T = 11$	2500	6000	10700	22700	28900	36800
$T = 17$	3500	11100	22300	31800	45700	62500
$T = 23$	7500	14900	26600	37800	55400	65800
$T = 41$	21300	29600	43400	60600	79200	103500

For all values examined, LLSO exhibits significant speedup compared to LFO. It is 12x faster for small problems ($T = 6$, $k = 0$), up to 1000x faster for large problems ($T = 41$, $k = 5$). For real-world problems, the values for T and k will be even higher, so speedup will be even more significant. The only difference between the $T=23$ and $T=41$ examples are rows made entirely out of zeroes, when all wells were turned off. This does not provide more information to the model and only slows down the solving process. To increase efficiency even further, rows should be removed from the y vector and A matrix if they contain only zeroes in both before solving the model. Because only a small amount of $m_{i,t}$ values are known (corresponding to points when $d_{i,t} = 1$), this will greatly reduce the size of the problem, making it much faster to solve. At the small scales of the provided simulated data, there did not appear to be a significant speedup from making this change, because of the computational cost associated with doing the deletions, However for larger

problems, this will be more efficient. The results produced by the original and row-reduced methods are identical to negligible machine error.

3.3 Applying Tikhonov Regularisation to LLSO

Next, Tikhonov regularisation is applied to the LLSO Model. More rows need to be added to A_1 and y to achieve this regularisation, making the matrix approximately three times taller. The rows attached to A_1 will contain n finite-difference stencil blocks along the diagonal to numerically differentiate $q = A_2x$ for each well. To find the first derivative, a forward finite difference stencil is used. Similarly, to find the second order equivalent, a central difference is used. These are defined in Eq. (B.1) and Eq. (B.2) in Appendix B. These newly formed matrices $A_{Tik,1}$ and $A_{Tik,2}$ can be attached to the original A_1 matrix to give a new matrix,

$$A_{1,Tik} = \begin{bmatrix} A_1 \\ A_{Tik,1} \\ A_{Tik,2} \end{bmatrix} \in \mathbb{R}^{T(n+1)+n(T-1)+n(T-2) \times nT}, \quad (3.3.1)$$

that can be used in conjunction with the updated output vector,

$$y_{Tik} = \begin{bmatrix} y \\ 0 \\ 0 \end{bmatrix} \in \mathbb{R}^{T(n+1)+n(T-1)+n(T-2)}, \quad (3.3.2)$$

to conduct Tikhonov-regularised LLSO. Substituting these expressions into Eq. (3.1.2) gives the Tikhonov-regularised LLSO model equation,

$$y_{Tik} \approx A_{1,Tik}A_2x. \quad (3.3.3)$$

3.4 Regularisation Parameter Tuning for LLSO

There are three parameters that can be tuned in order to get the best results from LLSO. To quantify the results, Root Mean Square Error (RMSE) was chosen. This was calculated across all wells and timepoints, as even the provided TFT points will not be met perfectly by LLSO. The parameter space $\mathbf{\Lambda} = [\lambda_1, \lambda_2, \lambda_3]$ where λ_1 represents the weighting put on the TFT points, λ_2 represents the weighting of the first derivative Tikhonov regularisation and λ_3 represents the weighting of the second derivative Tikhonov regularisation.

To conduct parameter tuning, a Particle Swarm Optimisation (PSO) metaheuristic was used. This is an optimisation algorithm inspired by swarming animals such as insects and fish [39]. This metaheuristic will give good (but not necessarily optimal) results in a reasonable amount of time and can be easily applied to a black-box function. A fixed value of $k = 5$ was used for testing. The $\hat{\mathbf{\Lambda}}$ that gave the lowest RMSE for the simulated dataset was $\hat{\lambda}_1 = 160.13$, $\hat{\lambda}_2 = 5.52$ and $\hat{\lambda}_3 = 0.00$. This resulted in a RMSE of 1.139 kg/s.

The fact that $\hat{\lambda}_3$ has an ‘optimal’ value of zero indicates that second derivative Tikhonov regularisation is unnecessary, and that first derivative Tikhonov regularisation is more effective. The fact that $\hat{\lambda}_1$ has an ‘optimal’ value of 160, which is greater than the ‘balanced’ value for λ_1 of 93, indicates that fitting the polynomial to the known TFT points is more important than fitting the total extraction to the bulk extraction data. In chapter 4, these optimal values $\hat{\mathbf{\Lambda}}$ will be used to produce final results, which will be discussed in detail in chapters 4 and 5 and compared with Michael’s GPR model.

Chapter 4

Results with Simulated and Real Data

This chapter will discuss the qualitative and quantitative results produced by the final LLSO model designed in chapter 3. The model's effectiveness on both simulated and real data will be discussed, and the uncertainty of its predictions will be quantified using Monte Carlo simulation uncertainty propagation.

4.1 Quantitative Results of LLSO on Simulated and real Data

The best results found from parameter tuning of the LLSO Model are shown in Fig. 4.1. This is clearly the best result yet, as the model nearly perfectly matches the true data. It is not overfit or underfit. The RMSE of 1.139 is the lowest for any model so far. This indicates that for simple scenarios such as this simulated dataset, LLSO is an effective tool for inferring the extraction history of a geothermal field. However real datasets are likely to be much more messy, with missing data and measurement errors meaning that data will never exactly add up.

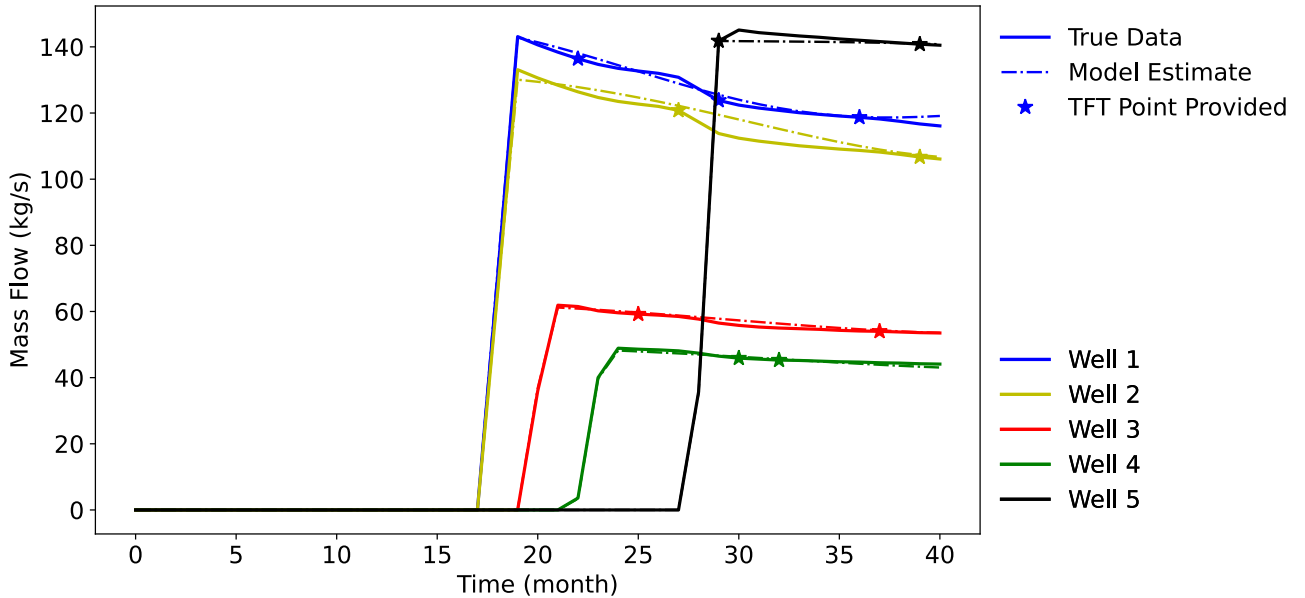


Figure 4.1: Mass flows predicted by LLSO for simulated data, for $k = 5$ and optimal parameter values found using particle swarm optimisation. The RMSE for this prediction is 1.139.

Real data taken from a New Zealand geothermal field was used for testing. Because this is

real data, the true values for mass flow are not known. However, an expert's guess as to how much mass was extracted from each well during each day has been provided. This is not necessarily an accurate measurement, but it is currently the best guess that exists, so it will be helpful to compare the model's results against this estimate. The TFT points are now separate from the 'true' data as that no longer exists, so have to be provided to the model separately. Eq. (3.1.4) is modified so that the m matrix now represents the provided TFT points, and the d matrix now represents whether or not a TFT point was or wasn't provided to the model. Similarly, the z matrix of when wells were on or off are now determined not from the 'true' data, but instead from the expert's estimate.

Michael and I manipulated the real data so that it could be inputted to our models. This was mainly achieved by simply manipulating the data within an Excel spreadsheet, however the provided TFT points required more intense data analysis before it was possible to pick out which points should be used where within the model. For this, an R script was used to analyse and manipulate the TFT data from the provided Excel sheet, removing missing values and appropriately matching the points to the correct day of measurement. From the available data, Separator 3 for a given time-period was chosen to test our models, because having only three active wells, and a timeframe of around 1000 days produced clearer plots that could easily be compared to each other's and the expert's estimate. The results produced by LLSO are shown in Fig. 4.2.

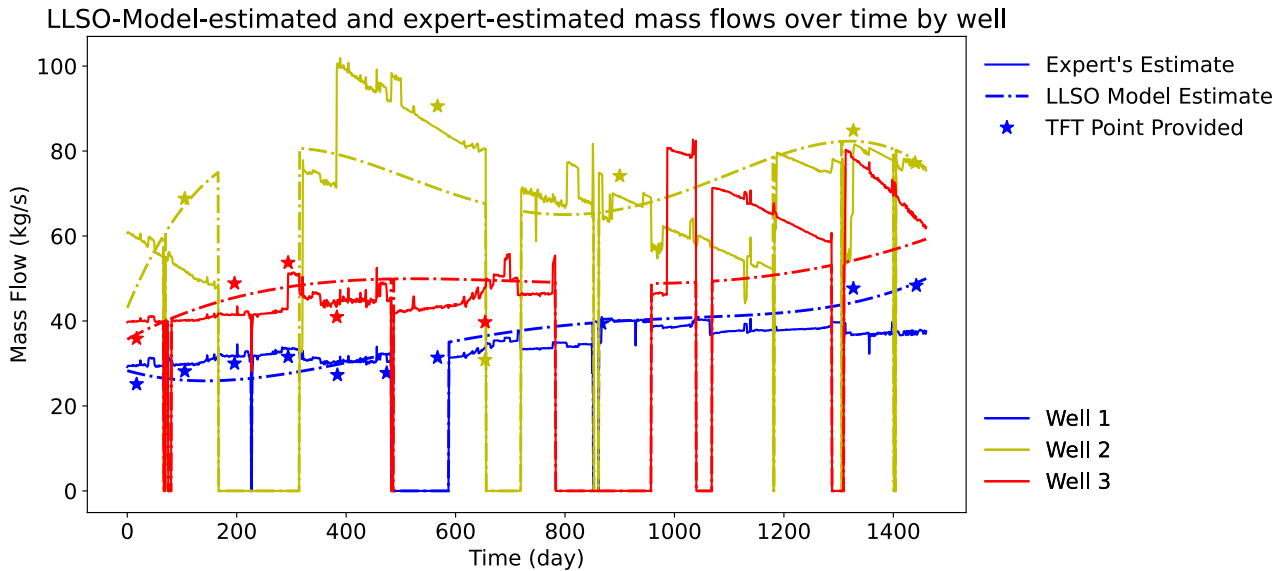


Figure 4.2: Mass flows predicted by LLSO for real data, for $k = 5$ with optimal regularisation parameter values. This somewhat follows the trend of the TFT points, similarly to the expert's estimate. However, it does differ in places, especially for Well 2 which varies the most sharply. It does best with Well 1 because Well 1 provides the most TFT points.

The model estimate shown in Fig. 4.2 somewhat aligns with the expert's estimate. For Well 1, which has many TFT points, the model estimate appears to be a better estimate than the expert's estimate because it more closely matches the provided TFT points. For Well 2, The estimates also match fairly closely throughout the full 1461 days. At certain points, the LLSO model under-predicts (relative to the expert's estimate), and at other points is over-predicts, so overall it predicts a similar amount of extraction. For Well 3, the estimates align closely through the first 800 days when there are many TFT points, but diverge after this point, when there are less TFT points provided. It is difficult to tell which estimate is more accurate in this region. RMSE cannot be computed against the true extraction from this well as it is not known, but it can be computed against the expert's estimate. However, this would be a poor measurement of the accuracy of the model as the expert's estimate could be at least as wrong as the model's estimate.

4.2 Quantifying LLSO Uncertainty

To quantify the uncertainty of the final model, Monte Carlo simulation uncertainty propagation was used. Gaussian noise with a mean of 0 and a standard deviation of 5.0 kg/s was added to the known TFT points, to represent the uncertainty in the provided data. 2000 models were then run, each getting a different output because of the added noise. This then allows the variance of the output model to be quantified. The standard deviation was calculated for each timepoint for each well, and was found to be in the range of 3.29 to 14.6 kg/s. The mean standard deviation across all points when a well was fully on was 5.47 kg/s. This was larger than the inputted standard deviation of 5.0 kg/s, so some additional uncertainty is being added by the LLSO Model, as expected. However, the model only increases the average variance by 19.6%, which is a small amount.

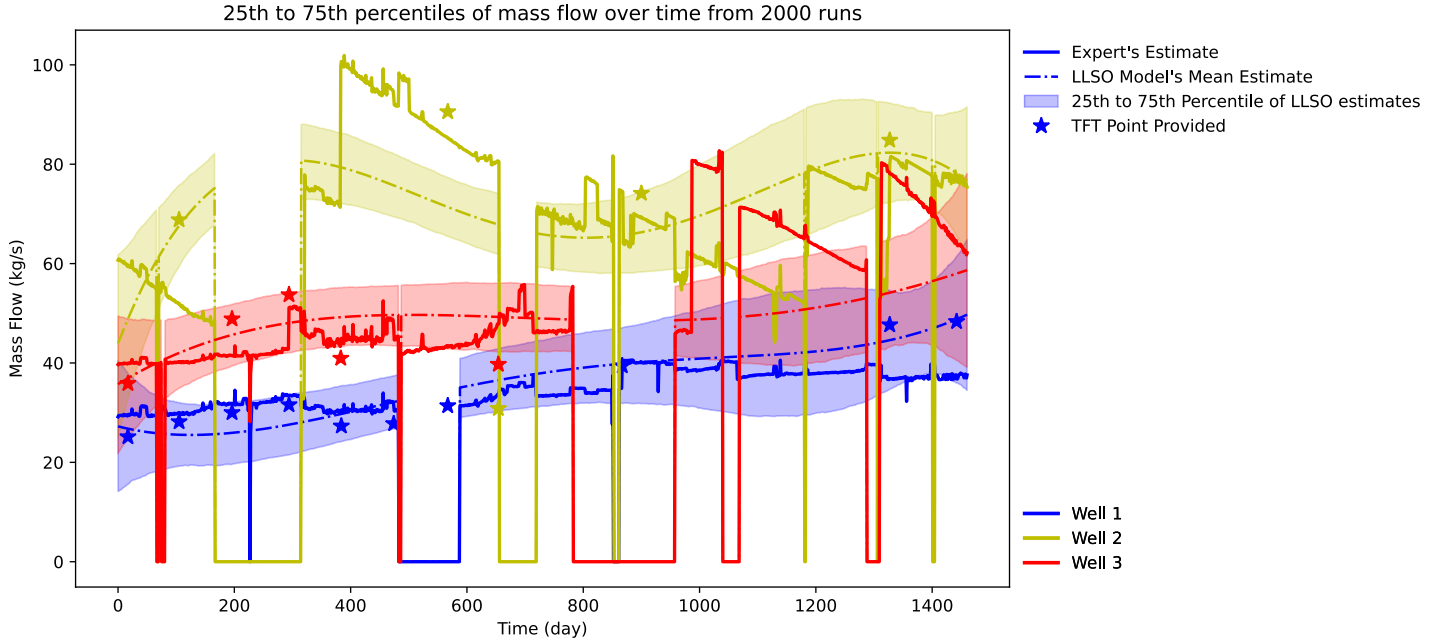


Figure 4.3: 25th to 75th percentiles of mass flows predicted by LLSO for real data, for $k = 5$ and optimal parameter values. These percentiles graphically show the uncertainty of the prediction for each well and timepoint. There is more uncertainty when TFT points are not provided nearby, and there is more uncertainty at the edged of the time-domain.

The 25th to 75th percentile of predictions are shown in Fig. 4.3. This shows how the uncertainty is lower near TFT points, but increases when further from TFT points. Similarly, the uncertainty increases near the edges of the time-domain when the model is effectively extrapolating instead of interpolating. While the LLSO does not provide highly certain estimates, it is important that uncertainty can be quantified, so that geothermal modelers can input this uncertainty into predictive models.

4.3 Comparison of LLSO and GPR Model Results

Michael's Gaussian Process Regression (GPR) Model produces quite different results from my LLSO model. These are shown in Fig. 4.4. Michael's results have more variability in predictions, and allow for sharp changes in mass flows for each well, whereas LLSO enforces smoothness, and therefore gives a more consistent estimate. Michael's model seems to predict less uncertainty, as the 95% confidence intervals shown converge tightly around known TFT points, for both the known well and the other wells. This is because GPR enforces each sample drawn to go exactly through each TFT point, with no uncertainty applied to that measurement. This can be considered to be overfitting to the data because this enforced certainty results in a highly variable prediction for each well over time, with sharp increases and decreases linking the known TFT points.

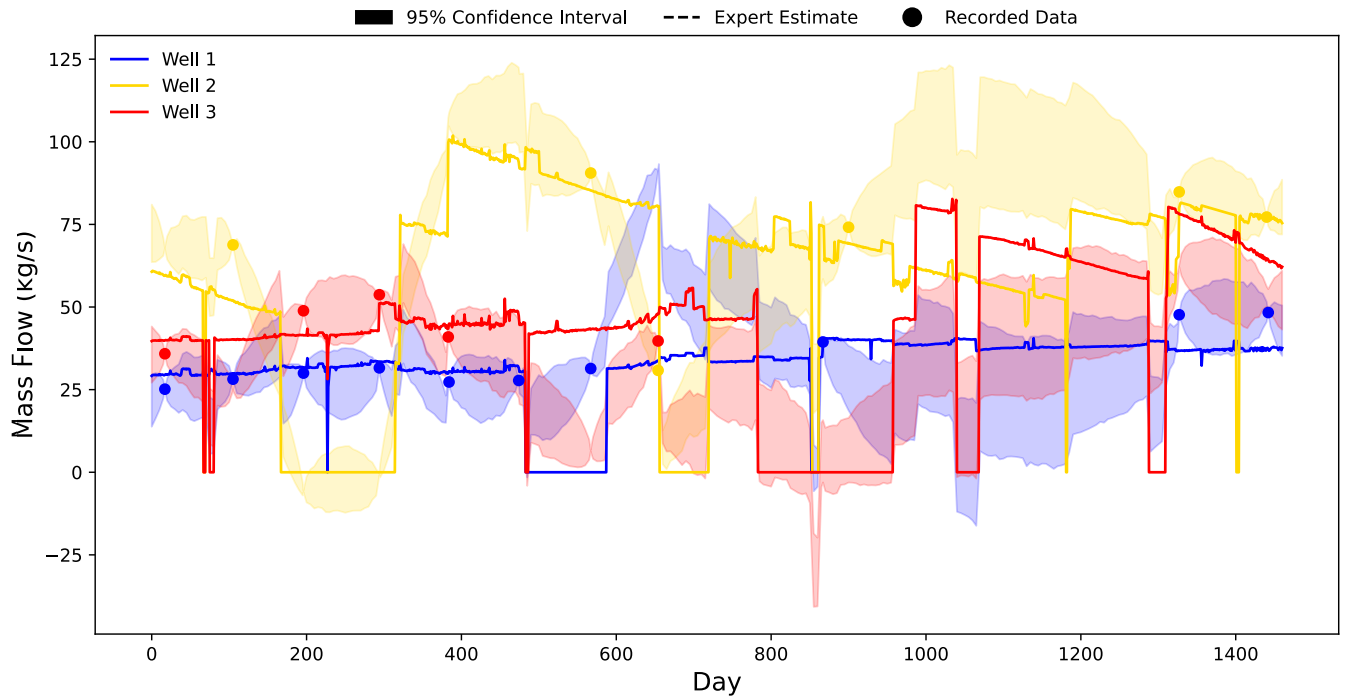


Figure 4.4: 95% confidence intervals of Michael’s GPR Model for real data. This shows a highly variable prediction because each sample is enforced to go exactly through each TFT point. Uncertainty increases quickly further from TFT points. Adapted from [36]

Chapter 5

Discussion

5.1 Discussion of LLSO and GPR Model Effectiveness

As a first attempt at predicting the data history of a geothermal field using data-driven methods, the LLSO and GPR models both succeed. Visually, the deterministic LLSO model appears to provide a better prediction for the data history of each well as it is smoother and more consistent. Meanwhile, the Bayesian GPR model produces results with more easily quantifiable uncertainty, and naturally produces an interval of predictions instead of a single prediction, which may be helpful for geothermal modelers.

By comparing Fig. 4.3 and Fig. 4.4, it can be seen that the LLSO model (using Monte Carlo simulations) predicts more uncertainty than the GPR model. This is likely because the LLSO predictions are not forced to exactly meet each TFT point, while the GPR predictions are. With more contextual knowledge of the uncertainty associated with these data points, it may be possible to determine which approach makes more sense. If there is high confidence in the TFT measurements then the GPR approach makes more sense, but if there is low confidence in these measurements (which is more likely for this specific dataset), the LLSO approach is more realistic. If this is not known, perhaps an ensemble method using results from both methods could also be useful.

The enforced smoothness (due to Tikhonov regularisation) of the LLSO model makes it produce more realistic results than the GPR model. For example, for well 1 the LLSO predicts a slowly increasing trend across time, meanwhile the GPR model predicts a highly varying extraction which goes from around 25 kg/s to around 90 kg/s within only 50 days. Also, the LLSO model appears to be more realistic because (while nothing explicitly prevents it from doing this) it never predicts any negative mass flows. Contrastingly, the GPR model predicts a negative mass flow at least once for each well. However, this enforced smoothness can also be over-restricting, resulting in LLSO struggling to model the finer behaviour of geothermal wells over time. This may not be a large issue, as geothermal modelers will be more concerned with the big-picture of the mass extraction, as opposed to the small variations that naturally occur over time. However, this is still a drawback of the LLSO model.

The LLSO model predictions for a given well also appear to be more independent of other well predictions. For example, when LLSO predicts a temporary increase in the extraction of well 2, there is not a (noticeable) corresponding decrease in wells 1 and 3. Contrastingly, GPR does predict a corresponding decrease because it enforces the three well measurements to add up exactly to the total extraction at each point. The LLSO model allows there to be some error here, which enables more independence between wells. This could be considered a strength or weakness, depending on whether this is expected behaviour for a given geothermal field. For example, if the wellhead pressures are intentionally adjusted so that when one well produces less another will take its place and produce more, then the GPR model results make sense. However, if one well decreases its output and there is no change to the other wells then the LLSO model results would model this more accurately. This assumption of perfect data could be relaxed in the future

for the GPR model, which may provide smoother predictions which are more similar to that of the LLSO model.

The accuracy of both models is limited by the amount and certainty of data that is provided to them. The rarity of TFT measurements, combined with their high uncertainty (at least for this dataset) are the main thing preventing the LLSO and GPR models from making better and more certain predictions. There are TFT measurements in the provided real dataset which are clearly wrong. Some values are higher than the total extraction mass on that day, which is impossible. Many other data points have notes from a reservoir engineer saying that the flow was too high, too low, or otherwise infeasible. The separator 3 data shown in Figures 4.3 and 4.4 was one of the cleanest parts of the overall dataset because it has reasonable TFT measurements and enough of them to make decent predictions. If the models were run on more noisy and unreliable subsets of the real data, they would produce less helpful predictions.

Available computational power is also a limiting factor for these models. Michael’s GPR model takes around 10 minutes to run on the Separator 3 data, so will be unable to handle larger datasets without either improved efficiency or significantly more computation power. This is similar to the LFO model, which reached it’s computational limits with the size of the simulated dataset. Contrastingly, the LLSO model is the most efficient of the models proposed, and has not yet reached its computational limits. It will need to be tested on larger datasets to determine where these limits are, if they exist at all. This is an important factor and makes LLSO a strong candidate to be used in industry.

Table 5.1: Summary table of RMSE values for different LFO and LLSO models. This shows the iterative improvement that occurred throughout the model development process.

Model Name	k	z	Regularisation	RMSE [kg/s]
Constant LFO	0	binary	none	10.36
Linear LFO	1	binary	none	10.34
Linear LFO with non-binary z	1	non-binary	none	1.220
AHOP LFO (cubic)	3	non-binary	none	13.24
Ridge-Regularised AHOP LFO	3	non-binary	Ridge	3.787
Tikhonov-Regularised AHOP LFO	3	non-binary	1st & 2nd order Tik.	1.333
Initial LLSO	5	non-binary	none	246300
Tikhonov-Regularised LLSO	5	non-binary	1st & 2nd order Tik.	1.362
Optimally-Regularised LLSO	5	non-binary	1st order Tik.	1.139

While the LLSO model shown in Fig. 4.2 is the best produced so far, it is also worth considering the iterative process that led to this point. Table 5.1 shows the RMSEs of each major iteration of both the LFO and LLSO models throughout this project, when tested on the same simulated dataset. This shows a clear improvement over time, with the final regularised AHOP LLSO model giving the lowest root mean square error thus far. It is logical to conclude that upon further iteration, it would be possible to decrease the RMSE even further, thus creating even better models. Some possible changes and improvements that could be made are outlined in the following section.

5.2 Future Work and Conclusions

Overall, both the LLSO and GPR models are a vast improvement over trying to estimate the extraction history manually. LLSO provides a reliable model which is efficient enough to handle large datasets with ease. Importantly, using the LLSO model removes the subjectivity of the manual process, and instead provides a robust and objective method for inferring the extraction history of a geothermal field. It has developed a lot over the course of this project but there is still room for improvement through future work. In

particular, there are five suggestions below for specific next steps that would improve the performance of the LLSO model even further:

- The most significant thing constraining the model (other than the rarity and uncertainty of data) is that it can only model the mass extraction history using polynomials. Even for high degree polynomials this restricts the behaviours that can be modeled by LLSO. This could be improved in a number of ways. Trigonometric terms such as $\sin(t)$ or $\sin(\frac{1}{365}t)$ could be added to the polynomial to incorporate periodic activity if there is known seasonal variation at a given geothermal field. Alternatively, the polynomial could be broken up into a number of splines which combine to form a continuous function across the whole time domain. This would also enable more sharp changes to occur as opposed to a single polynomial which can only change smoothly across time.
- Non-negativity should also be enforced for the LLSO model. While it has not made any negative predictions thus far, there is nothing preventing the model from making a negative mass flow prediction. This would be a large issue as this is clearly not possible. To fix this, LLSO could be performed iteratively. If it predicts any negativity, an artificial TFT point (or multiple) could be added to the data to ‘pull’ the prediction in the positive direction. This would be repeated until the model is making only non-negative predictions.
- The confidence in TFT values is often not constant between data points. This could be reflected by including weightings on TFT values which reflect the confidence in their accuracy. Weightings of 1 would indicate an average confidence in that point, while > 1 indicates high confidence and < 1 indicates low confidence. These would give higher importance to the model matching more certain TFT measurements, and would reduce the importance of matching the prediction to less certain measurements. This could also be incorporated into uncertainty quantification by adding noise with higher variance to TFT points which come with a lower level of certainty.
- The computational and data limits of the LLSO also need to be investigated further. The real data provided to the LLSO model in chapter 4 is a small subsection of the extraction history of that field. It would be important to test the model on more computationally intensive datasets to determine if it can be effectively used in an industry setting. Similarly, it will be important to test how much data is required to be fed to the model for it to work effectively, and whether there is a lower limit of TFT points, below which the model fails to produce reasonable results.
- Similarly, to enable the model to make predictions for more complicated geothermal systems, the model would need to be able to account for a number of complexities which are not yet accounted for. For example, there are cases when a well may initially be contributing to Separator A, but at some point switched to Separator B, or could even contribute to both Separators A and B simultaneously. These operational changes would have to be accounted for as an additional form of data input to the LLSO model.

In conclusion, the LLSO model successfully provides an objective and fast alternative to manually estimating the extraction history of a geothermal field. It makes reasonable predictions on both simulated and real data, and its uncertainty can be quantified using Monte Carlo simulation. However, it is limited by the quantity and accuracy of TFT points that it is provided, and can struggle to model the finer behaviour of the geothermal field because it only uses a single polynomial to model the whole extraction history for each well. Depending on the data provided, and the use case of the outputted data history, either the LLSO or GPR model may be preferred over the other, but for the real data provided the LLSO appears to model the extraction history more accurately. There are a number of next steps that can be taken to improve the LLSO model, including enforcing non-negativity, using more complicated non-polynomial forward models, and individually weighting TFT points. If these changes are made this method has the potential to become applicable for many geothermal fields across the world.

References

- [1] J. H. S. Wendell A Duffield, Geothermal Energy, Clean Power from the Earth's Heat, U.S. Department of the Interior, 2003.
- [2] G. U.-b. Janusz A. Kubiak a, Simulation of the effect of scale deposition on a geothermal turbine, *Geothermics* 31 (2002) 545–562. doi:10.1016/S0375-6505(02)00013-5.
- [3] Nz geothermal fields, <https://www.nzgeothermal.org.nz/geothermal-in-nz/nz-geothermal-fields/> (2024).
- [4] I. Donaldson, M. Grant, An estimate of the resource potential of new zealand geothermal fields for power generation, *Geothermics* 7 (1978) 243–252. doi:10.1016/0375-6505(78)90014-7.
- [5] L. Rybach, Geothermal sustainability, *Geo-Heat Centre Quarterly Bulletin* 28 (2007).
- [6] M. P. Hochstein, Assessment and modelling of geothermal reservoirs (small utilization schemes), *Geothermics* 17 (1988) 15–49. doi:10.1016/0375-6505(88)90004-1.
- [7] R. van de Schoot et al., Bayesian statistics and modelling, *Nature Reviews Methods Primers* 1 (2021) 26pp. doi:10.1038/s43586-020-00001-2.
- [8] N. D. Athens, J. K. Caers, A monte carlo-based framework for assessing the value of information and development risk in geothermal exploration, *Applied Energy* 256 (2019) 113932. doi:10.1016/j.apenergy.2019.113932.
- [9] A. B. Downey, *Think Bayes*, Green Tea Press, 2012.
- [10] P. L. Bonate, A brief introduction to monte carlo simulation, *Clinical Pharmacokinetics* 40 (2001) 15–22. doi:10.2165/00003088-200140010-00002.
- [11] T. G. Dietterich, Ensemble methods in machine learning, in: *Multiple Classifier Systems*, Springer Berlin Heidelberg, Berlin, Heidelberg, 2000, pp. 1–15.
- [12] K. Csilléry, M. G. Blum, O. E. Gaggiotti, O. François, Approximate bayesian computation (abc) in practice, *Trends in Ecology I& Evolution* 25 (2010) 410–418. doi:10.1016/j.tree.2010.04.001.
- [13] M. Sadegh, J. A. Vrugt, Approximate bayesian computation using markov chain monte carlo simulation: Dream(abc), *Water Resources Research* 50 (2014) 6767–6787. arXiv:<https://agupubs.onlinelibrary.wiley.com/doi/pdf/10.1002/2014WR015386>, doi:10.1002/2014WR015386.
- [14] C. Barnes, S. Filippi, M. e. a. Stumpf, Considerate approaches to constructing summary statistics for abc model selection, *Stat Comput* 22 (2012) 1181–1197. doi:10.1007/s11222-012-9335-7.
- [15] A. Bharti, Approximate bayesian computation with domain expert in the loop, <https://www.youtube.com/watch?v=Gx18PH0kiY8> (2023).
- [16] X. Shi, D. Jiang, W. Qian, Y. Liang, Application of the gaussian process regression method based on a combined kernel function in engine performance prediction, *ACS Omega* 7 (2022) 41732–41743. doi:10.1021/acsomega.2c05952.

- [17] E. R. Okoroafor, Machine learning in subsurface geothermal energy: Two decades in review, *Geothermics* 102 (2022) 102401. doi:10.1016/j.geothermics.2022.102401.
- [18] P. M. B. Abrasaldo, S. J. Zarrouk, A. W. Kempa-Liehr, A systematic review of data analytics applications in above-ground geothermal energy operations, *Renewable and Sustainable Energy Reviews* 189 (2024) 113998. doi:10.1016/j.rser.2023.113998.
- [19] M. I. Jordan, T. M. Mitchell, Machine learning: Trends, perspectives, and prospects, *Science* 349 (2015) 255–260. doi:10.1126/science.aaa8415.
- [20] Q. Wang, Y. Ma, K. e. a. Zhao, A comprehensive survey of loss functions in machine learning, *Annals of Data Science* 9 (2022) 187–212. doi:10.1007/s40745-020-00253-5.
- [21] G. H. Golub, P. C. Hansen, D. P. O’Leary, Tikhonov regularization and total least squares, *SIAM Journal on Matrix Analysis and Applications* 21 (1999) 185–194. doi:10.1137/S0895479897326432.
- [22] T. Hastie, Ridge regularization: An essential concept in data science, *Technometrics* 62 (2020) 426–433. doi:10.1080/00401706.2020.1791959.
- [23] M. R. Segal, Machine learning benchmarks and random forest regression, UCSF: Center for Bioinformatics and Molecular Biostatistics (2004). URL <https://escholarship.org/uc/item/35x3v9t4>
- [24] J. Li, L. Qiu, B. Tang, D. Chen, D. Zhao, R. Yan, Insufficient data can also rock! learning to converse using smaller data with augmentation, *Proceedings of the AAAI Conference on Artificial Intelligence* 33 (2019) 6698–6705. doi:10.1609/aaai.v33i01.33016698.
- [25] K. Poulinakis, D. Drikakis, I. W. Kokkinakis, S. M. Spottswood, Machine-learning methods on noisy and sparse data, *Mathematics* 11 (2023). doi:10.3390/math11010236.
- [26] C. Meng, S. Seo, D. Cao, S. Griesemer, Y. Liu, When physics meets machine learning: A survey of physics-informed machine learning (2022). arXiv:2203.16797. URL <https://arxiv.org/abs/2203.16797>
- [27] L. Yang, X. Meng, G. E. Karniadakis, B-pinns: Bayesian physics-informed neural networks for forward and inverse pde problems with noisy data, *Journal of Computational Physics* 425 (2021) 109913. doi:10.1016/j.jcp.2020.109913.
- [28] S. Cai, Z. Mao, Z. Wang, Physics-informed neural networks (PINNs) for fluid mechanics: a review, *Acta Mechanica Sinica* 37 (2021) 1727–1738. doi:10.1007/s10409-021-01148-1.
- [29] P. Banerjee, How AI and ML are changing simulation, <https://www.ansys.com/blog/how-ai-and-ml-are-changing-simulation> (2021).
- [30] Y. Song, A. Schwing, Richard, R. Urtasun, Training deep neural networks via direct loss minimization, in: M. F. Balcan, K. Q. Weinberger (Eds.), *Proceedings of The 33rd International Conference on Machine Learning*, Vol. 48 of *Proceedings of Machine Learning Research*, PMLR, New York, New York, USA, 2016, pp. 2169–2177.
- [31] S. Arridge, P. Maass, O. Öktem, C.-B. Schönlieb, Solving inverse problems using data-driven models, *Acta Numerica* 28 (2019) 1–174. doi:10.1017/S0962492919000059.
- [32] T. J. Dodwell, C. Ketelsen, R. Scheichl, A. L. Teckentrup, A hierarchical multilevel markov chain monte carlo algorithm with applications to uncertainty quantification in subsurface flow, *SIAM/ASA Journal on Uncertainty Quantification* 3 (2015) 1075–1108. doi:10.1137/130915005.

- [33] A. Mira, Mcmc methods to estimate bayesian parametric models, in: D. Dey, C. Rao (Eds.), *Bayesian Thinking*, Vol. 25 of *Handbook of Statistics*, Elsevier, 2005, pp. 415–436. doi:10.1016/S0169-7161(05)25014-9.
- [34] D. Rochman, W. Zwermann, S. C. van der Marck, A. J. Koning, H. Sjöstrand, P. Helgesson, B. Krzykacz-Hausmann, Efficient use of monte carlo: Uncertainty propagation, *Nuclear Science and Engineering* 177 (2014) 337–349. doi:10.13182/NSE13-32.
- [35] M. Cremon, M. Christie, M. Gerritsen, Monte carlo simulation for uncertainty quantification in reservoir simulation: A convergence study 2018 (2018) 1–18. doi:10.3997/2214-4609.201802226.
- [36] M. Terekhin, Inferring the detailed history of a geothermal field using sparse data and machine learning, BHons Research Project 83, Department of Engineering Science and Biomedical Engineering, The University of Auckland (2024).
- [37] Ardc nectar research cloud, <https://ardc.edu.au/services/ardc-nectar-research-cloud/> (2024).
- [38] Ardc jupyter notebook service, <https://ardc.edu.au/services/ardc-nectar-research-cloud/ardc-jupyter-notebook-service/> (2024).
- [39] D. Wang, L. Tan, Dapei Liu, Particle swarm optimization algorithm: an overview, *Soft Computing* 22 (2018) 387–407. doi:10.1007/s00500-016-2474-6.

Appendix A

Additional Figures

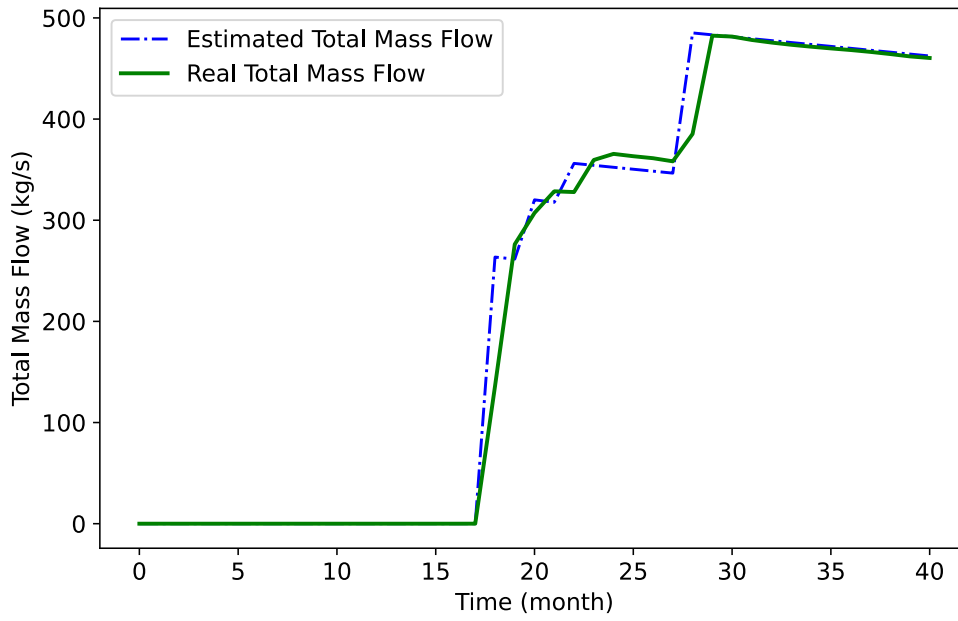


Figure A.1: Total mass flows predicted by linear LFO model against real results. This is an improved match than the constant model, because it is now a piecewise linear function, as opposed to piecewise constant function. This prediction has a RMSE of 10.36

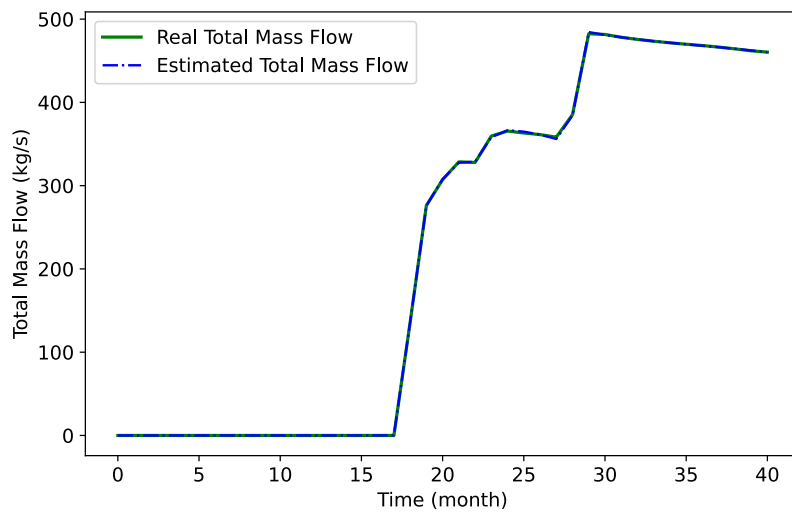
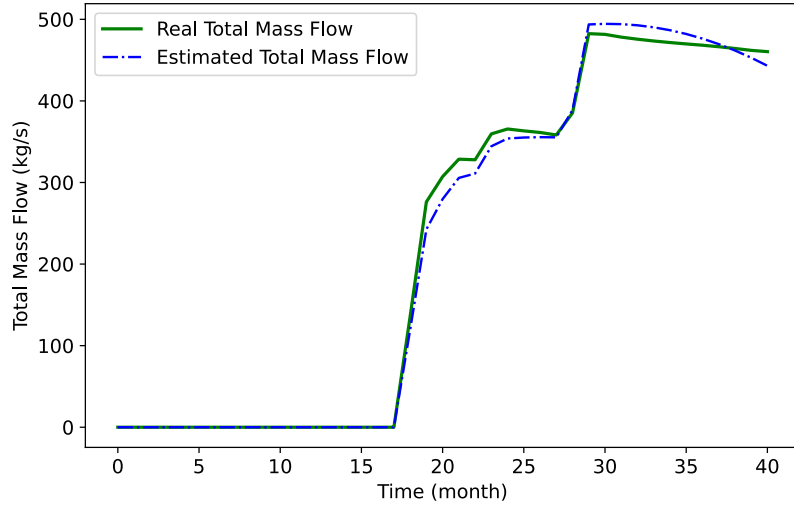
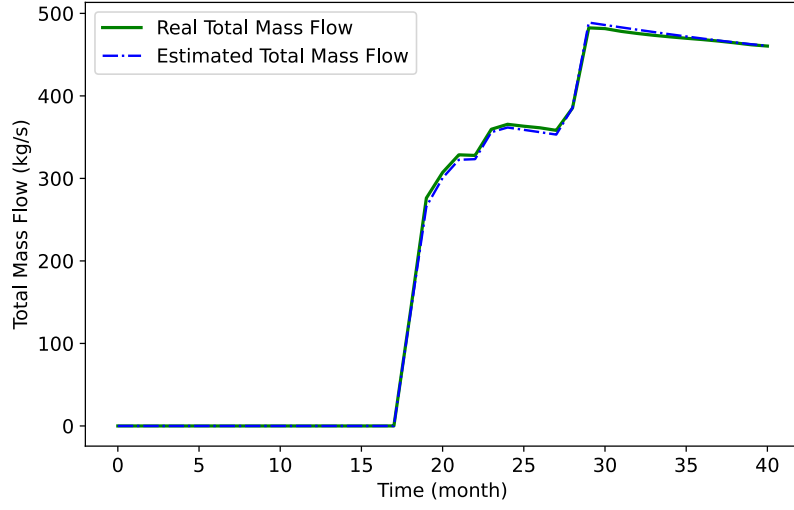


Figure A.2: Total mass flows predicted by unregularised cubic LFO model against true values. This produces an extremely overfit model, which predicts the true values nearly perfectly.



(A) Total estimated and real mass flows for ridge regularisation



(B) Total estimated and real mass flows for Tikhonov regularisation

Figure A.3: Total mass flows predicted by Cubic LFO Model under different regularisation schemes, against true values. Fig. (A) shows the results of Ridge regularisation, which is less overfit but also biases the prediction as discussed in section 2.6. Fig. (B) shows the results of first and second derivative Tikhonov regularisation which reduces overfitting while not biasing the solution, leading to a good prediction.

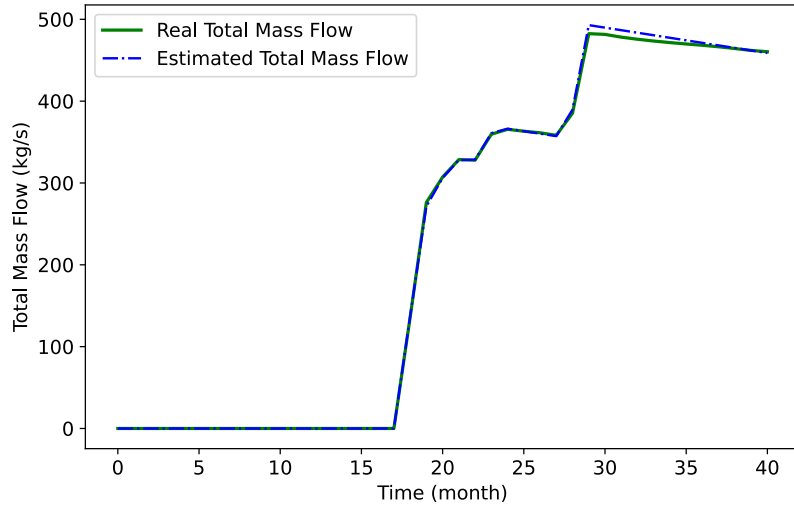


Figure A.4: Total mass flows predicted by LLSO model with optimal regularisation parameters. This uses only first-derivative Tikhonov regularisation, but produces a very close alignment to the true data.

Appendix B

Additional Equations

$$A_{Tik,1} = \sqrt{\lambda_2} \begin{bmatrix} -1 & 1 & 0 & \dots & 0 & 0 & \dots & \dots & \dots & 0 \\ 0 & -1 & 1 & 0 & \vdots & \vdots & & & & \vdots \\ \vdots & & \ddots & \ddots & \vdots & \vdots & & & & \vdots \\ 0 & \dots & 0 & -1 & 1 & 0 & \dots & \dots & \dots & 0 \\ 0 & \dots & \dots & \dots & 0 & -1 & 1 & 0 & \dots & 0 \\ \vdots & & & & \vdots & 0 & -1 & 1 & 0 & \vdots \\ \vdots & & & & \vdots & & & \ddots & \ddots & \vdots \\ 0 & \dots & \dots & \dots & 0 & 0 & \dots & 0 & -1 & 1 \end{bmatrix} \in \mathbb{R}^{n(T-1) \times nT} \quad (\text{B.1})$$

$$A_{Tik,2} = \sqrt{\lambda_3} \begin{bmatrix} 1 & -2 & 1 & \dots & \dots & 0 & 0 & \dots & \dots & \dots & \dots & 0 \\ 0 & 1 & -2 & 1 & & \vdots & \vdots & & & & & \vdots \\ \vdots & & \ddots & \ddots & \ddots & \vdots & \vdots & & & & & \vdots \\ 0 & \dots & \dots & 1 & -2 & 1 & 0 & \dots & \dots & \dots & \dots & 0 \\ 0 & \dots & \dots & \dots & \dots & 0 & 1 & -2 & 1 & \dots & \dots & 0 \\ \vdots & & & & & \vdots & 0 & 1 & -2 & 1 & & \vdots \\ \vdots & & & & & \vdots & & & \ddots & \ddots & \ddots & \vdots \\ 0 & \dots & \dots & \dots & \dots & 0 & 0 & \dots & 0 & 1 & -2 & 1 \end{bmatrix} \in \mathbb{R}^{n(T-2) \times nT} \quad (\text{B.2})$$